The bubble on the dumbbell as a quantics tensor network

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I. THE DUMBBELL FAMILY

The spectrum of the Green function is a segment of the real axis, and the Fermi function, or the Bose function for bosons, has its poles at the Matsubara frequencies, on the imaginary axis at distances from the origin set by the temperature. Between the segment and the poles the integrands are analytic. We take the contours from the level curves of a conformal map ω that sends a strip, periodic in the coordinate ϕ along it and with the coordinate λ across it, $\zeta = \phi + i\tau\lambda$, onto the plane cut along the segment and along the imaginary axis beyond the first poles: the line $\lambda = 0$ goes to the segment and the line $\lambda = 1$ to the imaginary axis beyond the first poles [Fig. 1(a)]. Every horizontal line in between goes to a closed curve C_λ around the segment that leaves out every pole, the contour at level λ [Fig. 1(b)]. At the imaginary axis the region is squeezed between the segment and the first poles, so the curves are narrow there and bulge on either side, like dumbbells.

On every contour ϕ is a periodic coordinate, so a function analytic in the region has Fourier coefficients in ϕ that decay exponentially, and the trapezoidal rule on a uniform grid in ϕ converges exponentially (Appendix G). Moving from one contour to another is a shift of the strip by $i\tau\Delta\lambda$, which multiplies the n-th Fourier coefficient by $e^{-\alpha n\Delta\lambda}$, with $\alpha = \pi\tau/4$ [Eq. (H1), Appendix H], so a change of level is diagonal in Fourier space. We consider a function with poles on a segment $[-E, E]$ of the real axis and the Fermi or Bose function, with poles at the Matsubara frequencies $i\omega_n = i(2n+1)\pi/\beta$ or $i\nu_m = 2\pi im/\beta$. The temperature fixes the map through the first poles, where the curve $\lambda = 1$ begins.

For a segment $[-E, E]$ and the inverse temperature β, the map is

$$\omega(\zeta) = E\kappa' \operatorname{sd}\Big(\frac{K}{2}(\zeta+1)\,\Big|\,\kappa^2\Big), \qquad \tau = \frac{2K'}{K}, \tag{1}$$

with $\kappa/\kappa' = E\beta/\pi$, $\kappa' = \sqrt{1-\kappa^2}$, $\operatorname{sd} = \operatorname{sn}/\operatorname{dn}$, and K and K' the complete elliptic integrals of the first kind with parameters κ^2 and κ'^2. We write $C_\lambda(E, \beta)$ for its level curves. As an example, Fig. 1 shows the dumbbell family for $E = 3$ and $\beta = \pi$, that is $\kappa = 3/\sqrt{10}$, at the levels $\lambda = 0.1, \ldots, 0.9$. This family avoids the poles at the fermionic Matsubara frequencies and encloses the segment $[-3, 3]$. For $\lambda \to 0$ it reduces to the segment of the real axis, and for $\lambda \to 1$ to the imaginary axis beyond the first poles $\pm i$.

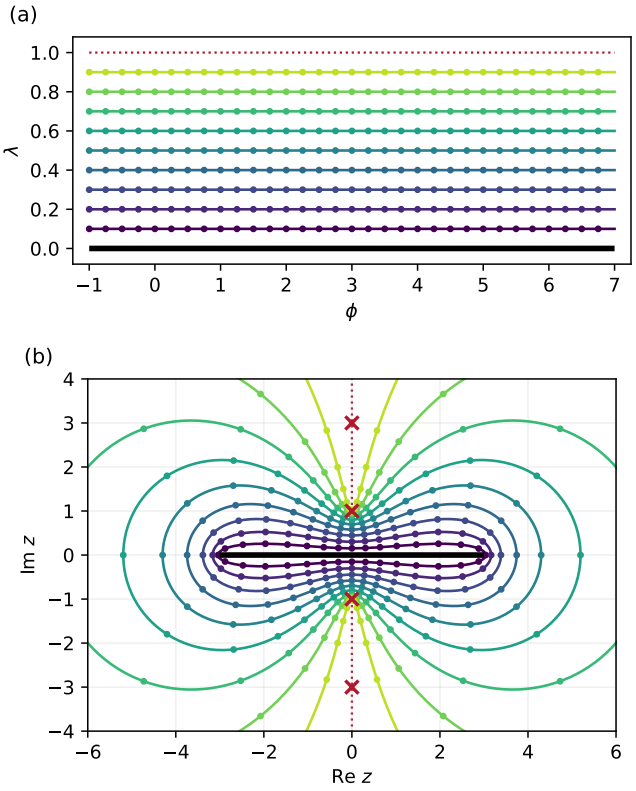


FIG. 1. The dumbbell family. (a) The strip $\zeta = \phi + i\tau\lambda$: the levels $\lambda = 0.1, \ldots, 0.9$ (colored lines) between the band ($\lambda = 0$, black) and the level $\lambda = 1$ (dotted) and the grid $\phi_j = -1 + 8j/N$ for $N = 32$ (dots). (b) Its image under ω: the dumbbells C_λ (same colors) around the band $[-3, 3]$ and the nodes $z_j = \omega_\lambda(\phi_j)$. The poles of the Fermi function at $\pm i, \pm 3i$ ($\beta = \pi$) lie on the level $\lambda = 1$, the imaginary axis beyond $\pm i$ (dotted).

A. Properties of the dumbbell map

We write $\omega_\lambda(\phi) = \omega(\phi + i\tau\lambda)$: ϕ is the position along a contour and λ its *level*, where $\tau = K'(k)/K(k)$, with $k = (L-1)/(L+1)$ and $L = \sqrt{1 + (E\beta/\pi)^2}$. The function sd satisfies $\operatorname{sd}(u + 2K) = -\operatorname{sd}(u)$, $\operatorname{sd}(u + 2iK') = -\operatorname{sd}(u)$ and $\operatorname{sd}(-u) = -\operatorname{sd}(u)$, and its Taylor coefficients are real, so

$$\begin{aligned} \omega(\zeta + 8) &= \omega(\zeta), & \omega(\zeta + 4) &= -\omega(\zeta), \\ \omega(2 - \zeta) &= \omega(\zeta), & \overline{\omega(\zeta)} &= \omega(\bar\zeta) = \omega(2 - \bar\zeta). \end{aligned} \tag{2}$$

In terms of ϕ and λ: period 8 in ϕ; $\omega_\lambda(\phi + 4) = -\omega_\lambda(\phi)$; and $\omega_\lambda(2 - \phi) = \overline{\omega_\lambda(\phi)}$, so the lower half of every contour

is the mirror image of its upper half.

For fixed $\lambda \in (0,1)$, $\phi \mapsto \omega_\lambda(\phi)$, $\phi \in [-1,7)$, traces a closed curve C_λ once, the *dumbbell at level* λ. Its shape follows from the values at $\lambda \to 0$ and $\lambda = 1$:

- $\lambda \to 0^+$: ω is real and runs $0 \to E \to 0 \to -E \to 0$ for $\phi = -1 \to 1 \to 3 \to 5 \to 7$. The contour collapses onto the segment $[-E, E]$: $\phi \in (-1,1) \cup (5,7)$ is the upper half ($\mathrm{Im}\,\omega > 0$), $\phi \in (1,5)$ the lower half. The *tips* are $\phi = 1$ and $\phi = 5$ ($\omega = \pm E$), and the curve crosses the imaginary axis at $\phi = -1 \equiv 7$ and $\phi = 3$ ($\omega = 0$). In increasing ϕ the curve is traversed clockwise.

- $\lambda = 1$: $\omega(\phi + i\tau)$ is purely imaginary with $|\mathrm{Im}\,\omega| \geq \pi/\beta$ ($\omega(-1+i\tau) = i\pi/\beta$, and $\omega \to \infty$ at $\phi = 1, 5$, where sd has its poles). The level-1 "contour" is the imaginary axis beyond the first poles, $\{\mathrm{Re}\,z = 0,\ |\mathrm{Im}\,z| \geq \pi/\beta\}$.

Every level $0 < \lambda < 1$ lies between the segment and the level-1 curve. The strip $0 < \lambda < 2$ is mapped by ω onto $\mathbb{C} \setminus [-E, E]$ (the level $2 - \lambda$ gives the mirror image $-\overline{C_\lambda}$ of C_λ, the same curve), and $\lambda = 2$ is the segment again.

The map is tied to the temperature through the poles of the Fermi function $f(z) = 1/(e^{\beta z}+1)$, at the fermionic Matsubara frequencies $i\omega_n = i(2n+1)\pi/\beta$ (Sec. II A). With $\kappa/\kappa' = E\beta/\pi$ the two closest to the real axis, $\pm i\pi/\beta$, sit at the ends of the level-1 curve, and then all poles lie on it. This is the choice of modulus of Lin, Lu, Ying and E, $L = \sqrt{1 + (E\beta/\pi)^2}$. Every contour C_λ with $0 < \lambda < 1$ separates the segment from all poles of f.

The Jacobian of the map follows from $v = \frac{K}{2}(\zeta + 1)$ and $\frac{d}{dv}\,\mathrm{sd}\,v = \mathrm{cn}\,v/\,\mathrm{dn}^2\,v$:

$$J(\zeta) \equiv \frac{\partial \omega}{\partial \phi} = E\kappa' \frac{K}{2} \frac{\mathrm{cn}\,v}{\mathrm{dn}^2\,v}, \qquad \frac{\partial \omega}{\partial \lambda} = i\tau\, J,$$
$$J^2 = \Big(\frac{K\kappa'\beta}{2\pi}\Big)^2 \Big(\frac{\pi^2}{\beta^2} + \omega^2\Big)(E^2 - \omega^2), \tag{3}$$

the second relation because ω is analytic in $\zeta = \phi + i\tau\lambda$, the third from $\mathrm{dn}^2\,v = 1/(1 + \beta^2\omega^2/\pi^2)$ and $\mathrm{cn}^2\,v = (1 - \omega^2/E^2)/(1 + \beta^2\omega^2/\pi^2)$. We write $J_\lambda(\phi) = J(\phi + i\tau\lambda)$. For $0 < \lambda < 1$, J_λ has no zeros on C_λ; on the segment it is $J_0(\varepsilon) = \frac{K\kappa'\beta}{2\pi}\sqrt{(E^2 - \varepsilon^2)(\pi^2/\beta^2 + \varepsilon^2)}$, which vanishes like $\sqrt{E - |\varepsilon|}$ at the edges, so the nodes cluster at the edges of the segment like Chebyshev points [Fig. 1(b)].

Every z off the segment and the level-1 curve lies on exactly one dumbbell $C_{\lambda'}$, and we call $\lambda(z) \equiv \lambda'$ the *level* of z.

C_λ crosses the imaginary axis at $\omega_\lambda(-1) = iE\kappa'\,\mathrm{sd}(K'\lambda\,|\,\kappa'^2)$ and $\omega_\lambda(3) = -\omega_\lambda(-1)$, by $\mathrm{sd}(iy\,|\,\kappa^2) = i\,\mathrm{sd}(y\,|\,\kappa'^2)$; these are the lowest point of its upper half and the highest point of its lower half. The highest and lowest points of C_λ follow from (3) (Fig. 2). Along C_λ the tangent J is horizontal where J^2 is real and positive. Since $J^2 \propto (\pi^2/\beta^2 + \omega^2)(E^2 - \omega^2)$, the imaginary part of J^2 for $\omega = x + iy$ is proportional to $2xy[E^2 - \pi^2/\beta^2 - 2(x^2 - y^2)]$, which off the axes vanishes on the hyperbola $x^2 - y^2 = c^2$, $c^2 = (E^2 - \pi^2/\beta^2)/2$,

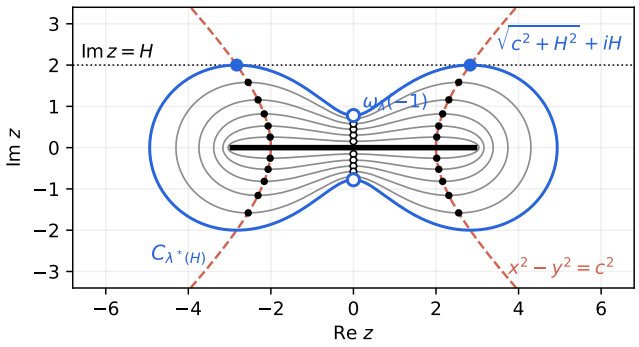


FIG. 2. Highest and lowest points of the dumbbells, for $E = 3$ and $\beta = \pi$. The extreme points (dots) of every C_λ (grey) lie on the hyperbola $x^2 - y^2 = c^2$ (dashed), where the tangent J is horizontal, and the open circles mark where they cross the imaginary axis. The dumbbell $C_{\lambda^*(H)}$ (blue) touches the line $\mathrm{Im}\,z = H$ (dotted) at its highest points $\pm\sqrt{c^2 + H^2} + iH$, here for $H = \nu_1$, and it crosses the imaginary axis at $\pm\omega_\lambda(-1) = \pm iE\kappa'\,\mathrm{sd}(K'\lambda\,|\,\kappa'^2)$ (blue open circles); every C_λ with $\lambda < \lambda^*(H)$ stays below the line.

where J^2 is positive. The highest and lowest points of a dumbbell of height H are therefore

$$z = \pm\sqrt{c^2 + H^2} \pm iH,$$

and since the dumbbells are nested, C_λ stays within $|\mathrm{Im}\,z| < H$ exactly when $\lambda < \lambda^*(H) \equiv \lambda\big(\sqrt{c^2 + H^2} + iH\big)$. We call H_λ the height of C_λ, so that its top is $\sqrt{c^2 + H_\lambda^2} + iH_\lambda$ and $\lambda^*(H_\lambda) = \lambda$.

The contour formula (18) on the nodes z_j needs G at the shifted points $z_j \pm w_k$, which lie on the translated copies $C_\lambda \pm w_k$ of the contour. Each shifted point has a level; the lowest one,

$$\lambda'_{\min}(\lambda, \mu) = \min_{j,k}\ \lambda\big(z_j \pm w_k\big), \tag{4}$$

measures how close the copies come to the segment ($\lambda'_{\min} > 0$) and whether they stay outside C_λ ($\lambda'_{\min} > \lambda$). By the symmetry $w_{k+N/2} = -w_k$ the sets $\{z_j + w_k\}$ and $\{z_j - w_k\}$ coincide. The copies come closest to the segment when w_k is where C_μ crosses the imaginary axis and z_j at the highest or lowest point of C_λ. Raising μ or lowering λ raises $\lambda'_{\min}$, and $\lambda'_{\min} > \lambda$ holds for μ above a threshold $\mu_{\mathrm{fwd}}(\lambda)$, roughly for $\lambda \lesssim \mu/2$.

II. THE FORMULA BEHIND THE NETWORK

A. The bubble

Momenta are on the grid $q = 2\pi m/N_q$, $m = 0, \ldots, N_q - 1$, and all momentum sums and shifts are modulo 2π. The temperature is $1/\beta$; the fermionic and bosonic Matsubara frequencies are

$$\omega_n = \frac{(2n+1)\pi}{\beta}, \qquad \nu_m = \frac{2\pi m}{\beta}, \qquad n, m \in \mathbb{Z}, \tag{5}$$

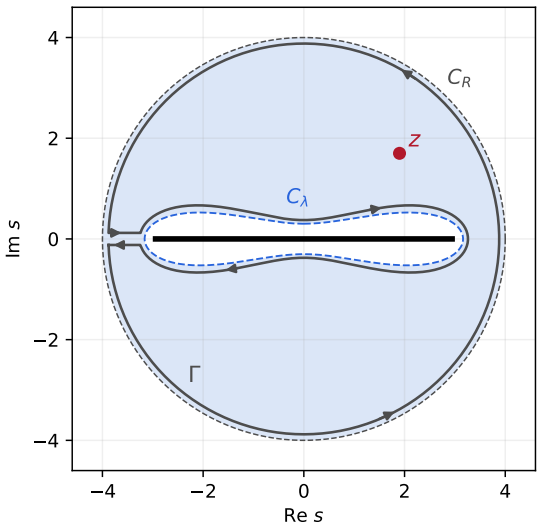


FIG. 3. Cauchy's integral formula, the first equality in (9), for G at a point z outside the dumbbell C_λ (dashed, $\lambda = 0.2$). G is analytic in the shaded region between C_λ and the circle C_R (dashed), and its singularities lie on the band $[-3,3]$ (black). A horizontal cut on the left, from C_λ to C_R, makes the region simply connected. Its counterclockwise boundary Γ (dark), drawn slightly inside it, is a single closed contour that encloses z and no singularity of G, so Cauchy's formula on it gives $G(z)$. The two sides of the cut are traversed in opposite directions and cancel, and Γ reduces to C_R counterclockwise and C_λ clockwise; the clockwise orientation turns $1/(s-z)$ into the $1/(z-s)$ of (9). As $R\to\infty$ the C_R integral vanishes.

and the Fermi function is

$$f(z) = \frac{1}{e^{\beta z}+1}, \tag{6a}$$

$$f(z+i\nu_m) = f(z), \tag{6b}$$

$$\operatorname*{Res}_{z=i\omega_n} f = -\frac{1}{\beta}. \tag{6c}$$

The one-particle Green function $G_q(z)$ is analytic off the band $[-E,E]$, and $G_q(z)\sim 1/z$ for $z\to\infty$. For any function ψ analytic inside and on C_λ, shrinking C_λ onto the band, where it runs along $\varepsilon - i0^+$ from left to right and back along $\varepsilon + i0^+$, gives

$$\frac{1}{2\pi i}\oint_{C_\lambda} ds\, G_q(s)\,\psi(s) = \int_{-E}^{E} d\varepsilon\, A_q(\varepsilon)\,\psi(\varepsilon), \tag{7}$$

with the spectral function

$$\begin{aligned} A_q(\varepsilon) &= \frac{G_q(\varepsilon - i0^+) - G_q(\varepsilon + i0^+)}{2\pi i} \\ &= -\frac{1}{\pi}\operatorname{Im} G_q(\varepsilon + i0^+), \end{aligned} \tag{8}$$

the jump of G_q across the band. For z outside C_λ, $\psi(s) = 1/(z-s)$ gives the spectral representation

$$G_q(z) = \frac{1}{2\pi i}\oint_{C_\lambda} ds\,\frac{G_q(s)}{z-s} = \int_{-E}^{E} d\varepsilon\,\frac{A_q(\varepsilon)}{z-\varepsilon}, \tag{9}$$

where the first equality is Cauchy's formula on the contour Γ of Fig. 3, which consists of a circle C_R, counterclockwise, and of C_λ, clockwise; the C_R integral vanishes as $R\to\infty$. With $\psi(s) = e^{-ist}$, (7) gives the time representation, the real-time Green function

$$\bar g_q(t) = \frac{1}{2\pi i}\oint_{C_\lambda} ds\, G_q(s)\, e^{-ist} = \int_{-E}^{E} d\varepsilon\, A_q(\varepsilon)\, e^{-i\varepsilon t}. \tag{10}$$

Conversely, since ε is real, $1/(z-\varepsilon)$ is a time integral: with $\sigma = \operatorname{sgn}\operatorname{Im} z$,

$$\frac{1}{z-\varepsilon} = -i\sigma\int_{-\infty}^{\infty} dt\,\theta(\sigma t)\, e^{i(z-\varepsilon)t}, \tag{11}$$

and inserting it into (9) gives G_q in terms of $\bar g_q$,

$$G_q(z) = -i\sigma\int_{-\infty}^{\infty} dt\,\theta(\sigma t)\, e^{izt}\,\bar g_q(t). \tag{12}$$

The integral converges for every z off the real axis, since $|\bar g_q(t)| \le \int d\varepsilon\,|A_q(\varepsilon)|$, and G_q is the Fourier transform of $-i\sigma\theta(\sigma t)\,\bar g_q(t)$, the retarded Green function for $\sigma = 1$ and the advanced one for $\sigma = -1$. With $\psi(s) = f(s)\,e^{ist}$, (7) gives the occupied real-time Green function $g_q(t)$, obtained from $\bar g_q(-t)$ by replacing G_q with fG_q and A_q with fA_q.

As an example we take a tight-binding chain with nearest-neighbour hopping,

$$\varepsilon_q = 3\cos q, \qquad G_q(z) = \frac{1}{z-\varepsilon_q}, \qquad A_q(\varepsilon) = \delta(\varepsilon - \varepsilon_q), \tag{13}$$

at the inverse temperature $\beta = \pi$; its band is $[-3,3]$, and the bubble has its cut on $[-6,6]$. For this model the bubble is the Lindhard function (F3), the reference for every accuracy quoted.

The bubble reads

$$\begin{aligned} \chi_Q(i\nu_m) &= \frac{1}{N_q\beta}\sum_{q,n} G_q(i\omega_n)\, G_{q+Q}(i\omega_n + i\nu_m) \\ &= \frac{1}{N_q}\sum_q \frac{1}{\beta}\sum_n h(i\omega_n), \end{aligned} \tag{14}$$

with

$$h(z) = G_q(z)\, G_{q+Q}(z + i\nu_m). \tag{15}$$

The Matsubara sum converges absolutely because the summand is $O(\omega_n^{-2})$. By (9) the poles of h, which merge into cuts for a continuous spectrum, lie on the band $[-E,E]$ of the real axis, from $G_q(z)$, and on the band moved down by ν_m, from $G_{q+Q}(z+i\nu_m)$. These lie at the heights 0 and $-\nu_m$, even multiples of π/β, while the poles $i\omega_n$ of f lie at odd multiples. So h is regular at the poles of f, and with the residue $-1/\beta$ of f in (6c), $\operatorname{Res}_{z=i\omega_n}[f(z)h(z)] = -h(i\omega_n)/\beta$. Hence

$$\frac{1}{\beta}\sum_n h(i\omega_n) = -\sum_n \operatorname*{Res}_{z=i\omega_n}\big[f(z)\,h(z)\big]. \tag{16}$$

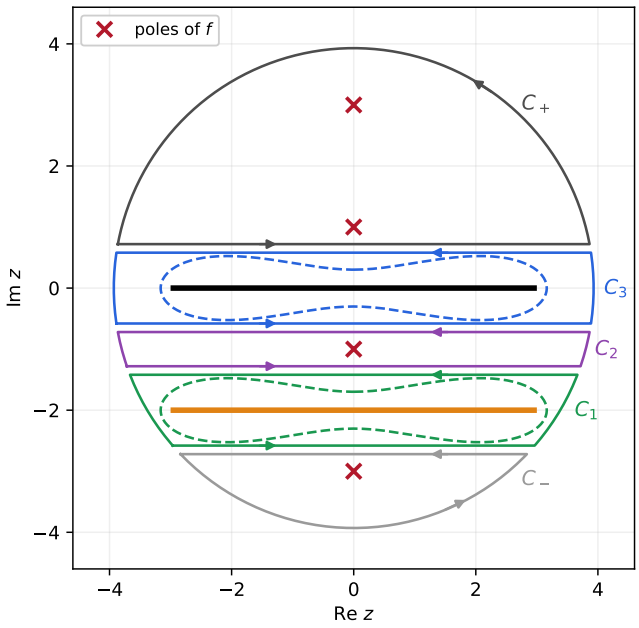


FIG. 4. The decomposition of the circle C_R that leads to (17), for $\beta = \pi$ and $m = 1$ ($\nu_1 = 2$). The poles of h of (15) lie on the band $[-3, 3]$ of the real axis (black), from $G_q(z)$, and on the band moved down by ν_m (orange), from $G_{q+Q}(z + i\nu_m)$. Horizontal lines at the distance η above and below them cut the disk into five regions, whose counterclockwise boundaries C_+, C_3, C_2, C_1 and C_- are drawn slightly inside them. C_3 encloses the band, C_1 the shifted band, and C_+, C_2, C_- the poles of f. Each line is traversed in both directions, so the five contours add up to C_R. The dashed curves are the dumbbells C_λ and $C_\lambda - i\nu_m$, here for $\lambda = 0.2$, into which C_3 and C_1 are deformed.

B. The contour formula

Let C_R be the counterclockwise circle centred at the origin with radius R, halfway between two poles of f. For $R \to \infty$ it contains both the poles of h, at $\mathrm{Im}\, z = 0$ and $\mathrm{Im}\, z = -\nu_m$, and those of f, and elsewhere inside it fh is analytic. For $m > 0$, as shown in Fig. 4, the contour integral over C_R can be rewritten as a sum of contour integrals over $C_R = C_- + C_1 + C_2 + C_3 + C_+$. These are the counterclockwise boundaries of the five regions into which horizontal lines, at the distance $\eta < \pi/\beta$ above and below the poles of h, cut the disk; each line is traversed twice, in opposite directions, and drops out. C_3 encloses the poles of h on the real axis and C_1 those on the shifted line, while C_-, C_2 and C_+ enclose only poles of f. As $R \to \infty$, the integrals of $fh/(2\pi i)$ over C_-, C_2 and C_+ give minus the right-hand side of (16), while the integral over C_R vanishes, because $h = O(z^{-2})$ and f stays bounded on

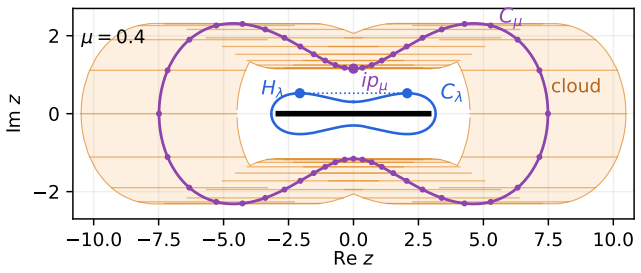


FIG. 5. The condition $\lambda < \Lambda(\mu, \beta)$ for $\lambda = 0.2$, $\mu = 0.4$ and $\beta = \pi$, with the vertical scale stretched 1.5 times. The thin lines are the shifted cuts $[-3, 3] \pm w$ for points w on C_μ (dots); for all w on C_μ they sweep the cloud $[-3, 3] \pm C_\mu$ (shaded), the zone that C_λ must avoid. Near the band its edge comes from the points $\pm ip_\mu$ where C_μ crosses the imaginary axis, and the dot on C_λ is its highest point. The point ip_μ lies above the highest point of C_λ, so C_λ lies in the hole of the cloud and (18) gives $\chi_Q(w_k)$.

C_R, so that

$$\sum_n \operatorname*{Res}_{z=i\omega_n} \big[f(z)\, h(z)\big] = -\frac{1}{2\pi i} \oint_{C_1} f\, h\, dz - \frac{1}{2\pi i} \oint_{C_3} f\, h\, dz. \tag{17}$$

The contour integrals over C_3 and C_1 can be deformed into integrals over the dumbbells C_λ and $C_\lambda - i\nu_m$ (dashed lines in Fig. 4) as long as C_λ leaves out the shifted band $[-E, E] - i\nu_m$. The dumbbell C_λ stays within $|\mathrm{Im}\, z| < \nu_m$ for $\lambda < \lambda^*(\nu_m)$ (Sec. I A), so the deformations are safe for $\lambda < \lambda^*(\nu_m)$, and since $\nu_m \geq \nu_1$, the bound for $m = 1$ covers every $m > 0$. By the change of variables $z \to z - i\nu_m$ in the integral over $C_\lambda - i\nu_m$ and by (6b), we have, at $w = i\nu_m$,

$$\chi_Q(w) = \frac{1}{N_q} \sum_q \frac{1}{2\pi i} \oint_{C_\lambda} dz\, f(z)\, G_q(z) \times \Big[G_{q+Q}(z + w) + G_{q-Q}(z - w)\Big]. \tag{18}$$

At $w = i\nu_m$ and for $\lambda < \lambda^*(\nu_m)$, the factors $f(z)\, G_{q\pm Q}(z \pm w)$ are analytic inside and on C_λ. Inserting (9), Cauchy's formula turns (18) into

$$\chi_Q(w) = \frac{1}{N_q} \sum_q \int_{-E}^{E} d\varepsilon \int_{-E}^{E} d\varepsilon'\, A_q(\varepsilon)\, A_{q+Q}(\varepsilon') \times \frac{f(\varepsilon) - f(\varepsilon')}{w + \varepsilon - \varepsilon'}, \tag{19}$$

with $w = i\nu_m$. Equation (19) does not depend on λ, is analytic in w off the interval $[-2E, 2E]$ of the differences $\varepsilon' - \varepsilon$ and vanishes as $|w| \to \infty$, so it is the analytic continuation $\chi_Q(w)$ of the bubble.

Now let w be a general complex number. The same steps give (19) as long as C_λ leaves out the shifted cuts

$[-E,E]\mp w$, where $G_{q\pm Q}(z\pm w)$ is singular. For $w=w_k\in C_\mu$, where $C_\mu=C_\mu(2E,\beta/2)=2\,C_\mu(E,\beta)$ is a dumbbell which encloses $[-2E,2E]$ and leaves out the poles $i\nu_m$, $m\neq 0$, of the Bose function, the shifted cuts sweep the set $[-E,E]\pm C_\mu$, the "cloud" in Fig. 5, so (18) holds on C_μ when the cloud lies outside C_λ. This happens when the top of C_λ lies below the point ip_μ where C_μ crosses the imaginary axis, with $p_\mu=2\,\mathrm{Im}\,\omega(-1+i\tau\mu)$ (Sec. I A):

$$H_\lambda < p_\mu = 2E\kappa'\ \mathrm{sd}(K'\mu\,|\,\kappa'^2).$$

In terms of levels this reads $\lambda<\Lambda(\mu,\beta)$, where $\Lambda(\mu,\beta)=\lambda^*(p_\mu)=\lambda\big(\sqrt{c^2+p_\mu^2}+ip_\mu\big)$. Since $p_\mu<\nu_1\le\nu_m$ and the level $\lambda^*(H)$ grows with H, $\Lambda(\mu,\beta)=\lambda^*(p_\mu)<\lambda^*(\nu_m)$, and the only condition on λ becomes $\lambda<\Lambda(\mu,\beta)$.

C. The bubble as a product in time

The data are the values of G on C_λ, while (18) needs $G_{q\pm Q}$ at the shifted points $z\pm w$, off the contour. The spectral form (19) is a bilinear form in the spectral functions with the kernel

$$\Psi_w(\varepsilon,\varepsilon')=\frac{f(\varepsilon)-f(\varepsilon')}{w+\varepsilon-\varepsilon'},\tag{20}$$

and (7), applied in each variable, gives

$$\begin{aligned}\frac{1}{(2\pi i)^2}\oint_{C_\lambda}dz\oint_{C_\lambda}ds\,G_q(z)\,G_{q'}(s)\,\Psi(z,s)\\ =\int_{-E}^{E}d\varepsilon\int_{-E}^{E}d\varepsilon'\,A_q(\varepsilon)\,A_{q'}(\varepsilon')\,\Psi(\varepsilon,\varepsilon')\end{aligned}\tag{21}$$

for every Ψ analytic in both variables inside and on C_λ. The right side depends on Ψ only through its values on the band, and with $\Psi=\Psi_w$ and $q'=q+Q$, summed over q and divided by N_q, it is the bubble (19). The Cauchy route of Appendix A applies (21) to Ψ_w itself, which requires Ψ_w to be analytic inside C_λ, that is, the copies $C_\lambda\pm w$ outside C_λ, $\lambda'_{\min}>\lambda$ in (4); on the nodes Ψ_w is then a dense tensor in z_j, z_m and w_k, compressed in Appendix A 3. Appendix A 4 obtains the compressed tensor from the time route, where this condition is not needed. The time route first writes Ψ_w on the band as an integral over time of products of exponentials, one factor per variable, and applies (21) to each product, which is analytic inside C_λ for every time.

For complex x and a direction γ in the complex t plane,

$$\frac{1}{x}=-i\int_0^{e^{i\gamma}\infty}dt\,e^{ixt},\qquad \mathrm{Im}(xe^{i\gamma})>0,\tag{22}$$

where t runs from 0 to infinity along the half-line $\arg t=\gamma$; for $\gamma=0$ and $\gamma=\pi$ this is (11). With $x=w+\varepsilon-\varepsilon'$ the integrand $e^{iwt}\,e^{i\varepsilon t}\,e^{-i\varepsilon' t}$ has one factor per variable, so in time the kernel is diagonal, a delta in t, which is the copy tensor of Fig. 6. Inserted into (19), (22) makes the integrals over ε and ε' factorize into $\bar g_q$ of (10) and g_q at complex t, where their contour forms still hold. The term with $f(\varepsilon)$ gives $g_q(t)\,\bar g_{q+Q}(t)$, and the term with $f(\varepsilon')$ gives $\bar g_q(-t)\,g_{q+Q}(-t)$, so the bubble becomes a product in time,

$$\begin{aligned}\chi_Q(w)=&-\frac{i}{N_q}\sum_q\int_0^{e^{i\gamma}\infty}dt\,e^{iwt}\\ &\times\big[g_q(t)\,\bar g_{q+Q}(t)-\bar g_q(-t)\,g_{q+Q}(-t)\big],\end{aligned}\tag{23}$$

exact when $\mathrm{Im}(we^{i\gamma})>2E|\sin\gamma|$, which makes $\mathrm{Im}(xe^{i\gamma})>0$ for all ε and ε' in the band. For real t, $\bar g_q(-t)=\bar g_q(t)^*$ and $g_q(-t)=g_q(t)^*$, since A_q and f are real on the band. The functions $\bar g_q$ and g_q are integrals of the data over C_λ, so G is never needed off the contour.

On C_λ the exponentials $e^{\pm ist}$ reach $e^{H_\lambda(\gamma)|t|}$, where $H_\lambda(\gamma)=\max_{s\in C_\lambda}\mathrm{Im}(se^{i\gamma})$ is the height of C_λ in the direction $ie^{-i\gamma}$, while $\bar g_q$ and g_q grow only like $e^{E|\sin\gamma||t|}$, as on the band. The node sums for them therefore carry rounding errors of relative size $e^{(H_\lambda(\gamma)-E|\sin\gamma|)|t|}$ times the machine precision, and the factor e^{iwt} damps these errors when

$$\mathrm{Im}(we^{i\gamma})>H_\lambda(\gamma)+E|\sin\gamma|,\tag{24}$$

that is, when w lies beyond every point $\varepsilon+s$, with $\varepsilon\in[-E,E]$ and $s\in C_\lambda$, in the direction $ie^{-i\gamma}$. We call the difference of the two sides the margin. Real time, $\gamma=0$ and $\gamma=\pi$, serves the upper and lower halves of C_μ, where (24) reads $\mathrm{Im}\,w>H_\lambda$ and $\mathrm{Im}\,w<-H_\lambda$. Imaginary time, $\gamma=\pi/2$ and $\gamma=3\pi/2$, serves the regions around the right and left tips, where it reads $\mathrm{Re}\,w>E+\omega_\lambda(1)$ and $\mathrm{Re}\,w<-E-\omega_\lambda(1)$, with $\omega_\lambda(1)=E\,\mathrm{nc}(K'\lambda\,|\,\kappa'^2)$, $\mathrm{nc}=1/\mathrm{cn}$, the tip of C_λ on the positive real axis. At the point ip_μ the margin of real time is $p_\mu-H_\lambda$, positive under the condition $\lambda<\Lambda(\mu,\beta)$ of (18). The Cauchy route needs the stronger condition $\lambda'_{\min}>\lambda$.

No single direction serves all of C_μ, and neither does any discretization of (23) whose times and weights do not depend on w. Such a discretization is an entire function $g(w)$, while χ_Q has its cut inside C_μ. With C_μ traversed counterclockwise, $\oint_{C_\mu}dw\,w\,g(w)=0$, while $\oint_{C_\mu}dw\,w\,\chi_Q(w)=2\pi i\,c_2$, where

$$\begin{aligned}c_2=&\frac{1}{N_q}\sum_q\int_{-E}^{E}d\varepsilon\int_{-E}^{E}d\varepsilon'\,A_q(\varepsilon)\,A_{q+Q}(\varepsilon')\\ &\times\big[f(\varepsilon)-f(\varepsilon')\big](\varepsilon'-\varepsilon)\end{aligned}\tag{25}$$

is the coefficient of w^{-2} in the expansion of (19) at large w. Since f decreases and $A_q\ge 0$, $c_2>0$ unless χ_Q vanishes, and

$$\max_{w\in C_\mu}|\chi_Q(w)-g(w)|\ge\frac{2\pi c_2}{L_\mu R_\mu},\tag{26}$$

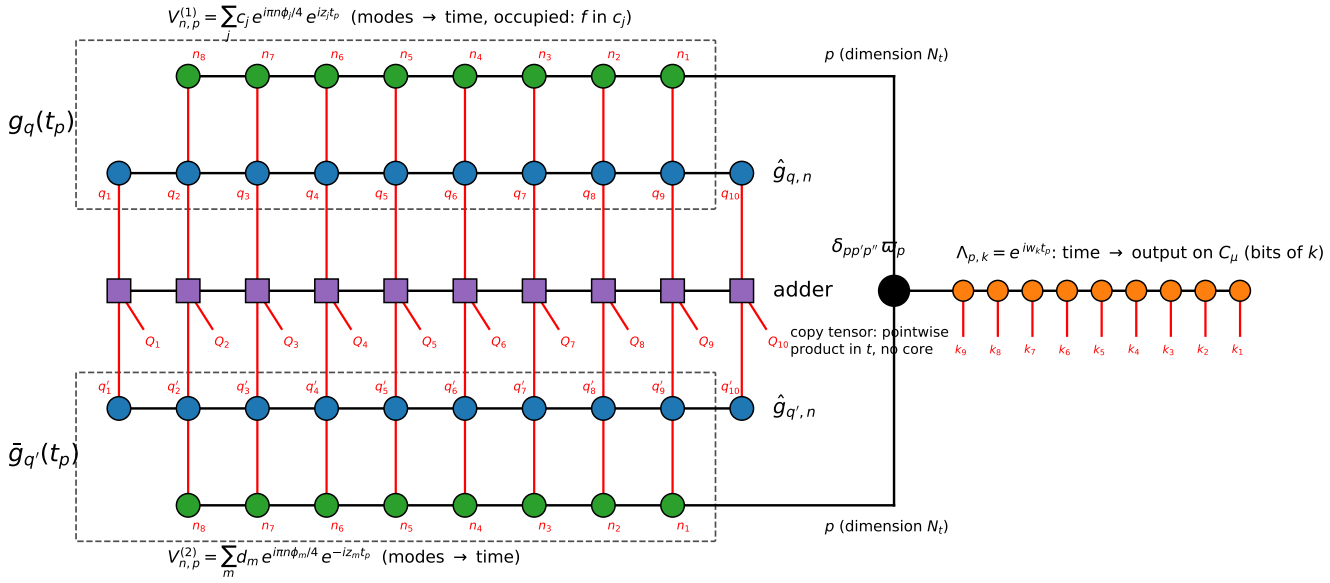


FIG. 6. The time-domain network, in the conventions of Fig. 12 (red legs are contracted, bits most significant first). Upper branch: the input train $\hat{g}_{q,n}$ (blue), its n legs contracted with the train $V^{(1)}_{n,p}$ (green), which carries the modes to the time points t_p and contains the Fermi function through c_j; the result is $A_{q,p}$, the function $g_q(t_p)$ in the gauge of (33), a train over the bits of q with the time leg p. Lower branch: the same with $V^{(2)}$ (weights d_m, no f, opposite sign in the exponent), which gives $B_{q',p}$, the function $\bar{g}_{q'}(t_p)$. The adder (purple) sums q and q' with the transfer momentum Q open. The two time legs meet at a copy tensor carrying the weights ϖ_p, the product in time, and the Laplace train $\Lambda_{p,k}$ (orange) sends the time leg to the output points of C_μ (bits of k). The time leg runs over the rules of the four directions; the labels give the leading factors of the tensors (33), without the gauge and the partition. The network gives $X_Q(w_k)$, and $\chi_Q(w_k)$ of (34) is $X_Q(w_k)$ plus the same train read at $-Q$ and $k + N/2$.

with L_μ the length of C_μ and $R_\mu = \max_{C_\mu}|w|$. The weights must therefore depend on the output point, and the direction of the time integral is the cheapest such dependence.

Let X_Q be the part of the bubble with $f(\varepsilon)$,

$$X_Q(w) = \frac{1}{N_q}\sum_q \int_{-E}^{E} d\varepsilon \int_{-E}^{E} d\varepsilon'\, \frac{f(\varepsilon)\, A_q(\varepsilon)\, A_{q+Q}(\varepsilon')}{w + \varepsilon - \varepsilon'}. \tag{27}$$

In the term of (19) with $f(\varepsilon')$, the substitution $q \to q - Q$ together with $\varepsilon \leftrightarrow \varepsilon'$ gives $X_{-Q}(-w)$, so $\chi_Q(w) = X_Q(w) + X_{-Q}(-w)$. For every direction with $\mathrm{Im}(we^{i\gamma}) > 2E|\sin\gamma|$, (22) turns (27) into $X_Q(w) = -i\int_0^{e^{i\gamma}\infty} dt\, e^{iwt}\, Y_Q(t)$, with $Y_Q(t) = N_q^{-1}\sum_q g_q(t)\,\bar{g}_{q+Q}(t)$; (23) is this formula applied to both terms with the same direction. Let $P_\gamma(w)$, $\gamma \in \{0, \pi/2, \pi, 3\pi/2\}$, be a partition of unity on C_μ, $\sum_\gamma P_\gamma(w) = 1$, with $P_\gamma(w) = 0$ wherever the margin of γ is below a fixed $m_0 > 0$; it exists when every point of C_μ has a direction with a margin of at least m_0. Then

$$\begin{aligned} \chi_Q(w) &= X_Q(w) + X_{-Q}(-w), \\ X_Q(w) &= -i\sum_\gamma P_\gamma(w) \int_0^{e^{i\gamma}\infty} dt\, e^{iwt}\, Y_Q(t), \end{aligned} \tag{28}$$

at every point of C_μ, with one fixed set of directions. Each direction is used only where (22) converges, so (28) is exact, and only where its margin is at least m_0, so the rounding error of each term is of the order of the machine precision divided by m_0. By (27), X_Q is analytic off $[-2E, 2E]$, like χ_Q, and every direction gives the same function where it is used, so the partition changes only how X_Q is computed.

D. The time domain on the nodes

The time functions g_q and $\bar{g}_q$ are integrals over C_λ. Parametrize C_λ by ϕ: $z = \omega_\lambda(\phi)$, $dz = J_\lambda(\phi)\, d\phi$. The parametrization is clockwise, so for any F analytic near the contour

$$\frac{1}{2\pi i}\oint_{C_\lambda} F(z)\, dz = -\frac{1}{2\pi i}\int_{-1}^{7} d\phi\; J_\lambda(\phi)\, F\big(\omega_\lambda(\phi)\big). \tag{29}$$

The integrand is 8-periodic in ϕ, and the trapezoidal rule on the uniform grid $\phi_j = -1 + j\,\Delta\phi$, $j = 0, \ldots, N-1$, with $\Delta\phi = 8/N$, is $\Delta\phi\sum_j$. The nodes are $z_j = \omega_\lambda(\phi_j)$, and the bubble is computed at the points $w_k = 2\,\omega(\phi_k + i\tau\mu)$ of C_μ on the same grid. With the Fermi function in

the integrand it gives

$$\frac{1}{2\pi i}\oint_{C_\lambda} f\,F\,dz \approx \sum_{j=0}^{N-1} c_j\,F(z_j),$$
$$c_j = d_j\,f(z_j), \qquad d_j = -\frac{\Delta\phi}{2\pi i}J_\lambda(\phi_j). \tag{30}$$

For every F that is analytic outside C_λ and $O(z^{-2})$ at infinity, the residue theorem between C_λ and a large circle, as in Sec. II B, gives $\sum_j c_j F(z_j) \approx \beta^{-1}\sum_n F(i\omega_n)$: the N nodes with the weights c_j replace the Matsubara frequencies. Without the Fermi function the rule reads $\frac{1}{2\pi i}\oint_{C_\lambda} F\,dz \approx \sum_m d_m\,F(z_m)$. The c_j depend on j, λ and β only, never on w, q or G, and the Fermi function is evaluated only at the nodes of C_λ. Both sets of weights are smooth in ϕ_j (rank 6 to 8 as quantics tensor trains, Appendix B). The quadrature error falls exponentially with N, like $e^{-\alpha N d}$ (G2), where d is the distance in level from C_λ to the nearest singularity of the integrand (Appendix G).

The two rules applied to g_q and $\bar g_q$ give the time functions from the data, at real or complex t,

$$g_q(t) \approx \sum_j c_j\,G_q(z_j)\,e^{iz_j t},$$
$$\bar g_q(t) \approx \sum_m d_m\,G_q(z_m)\,e^{-iz_m t}. \tag{31}$$

For each of the four directions γ_a the integral over $\tau = |t|$ in (28) is replaced by a rule $\int_0^{T_a} d\tau\,h(\tau) \approx \sum_p \varpi_p\,h(\tau_p)$. At the points where $P_{\gamma_a}(w_k) > 0$ the integrand decays at least like $e^{-d_a\tau}$, with d_a the smallest value of $\mathrm{Im}(w_k e^{i\gamma_a}) - 2E|\sin\gamma_a|$ there, and T_a is the time at which $e^{-d_a T_a}$ reaches the target accuracy. In real time the integrand oscillates with frequencies up to $\max_k|\mathrm{Re}\,w_k| + 2E$, which Gauss-Legendre panels of equal length resolve. In imaginary time it is a sum of exponentials with rates between d_a and $d_a + 4E$, and the panels double in length from the origin. The time leg p runs over the four rules, with the times $t_p = e^{i\gamma_p}\tau_p$.

A function given on the N nodes of C_λ has the discrete Fourier representation

$$G_q(z_j) = \sum_n \hat g_{q,n}\,e^{i\pi n\phi_j/4},$$
$$\hat g_{q,n} = \frac{1}{N}\sum_{j=0}^{N-1} G_q(z_j)\,e^{-i\pi n\phi_j/4}, \tag{32}$$

where n runs over any N consecutive integers (we use the two's-complement range $-N/2 \le n < N/2$); both relations are exact on the grid. Since $e^{i\pi n\phi_j/4} = e^{-i\pi n/4}e^{2\pi i n j/N}$, the coefficient $\hat g_{q,n}$ is $e^{i\pi n/4}/N$ times the standard FFT of the data.

In these modes the time functions are linear in the data. With a gauge h_p, for instance $h_p = H_\lambda(\gamma_p)$, the network is built from the tensors

$$V^{(1)}_{n,p} = \sum_j c_j\,e^{i\pi n\phi_j/4}\,e^{iz_j t_p - h_p\tau_p},$$
$$V^{(2)}_{n,p} = \sum_m d_m\,e^{i\pi n\phi_m/4}\,e^{-iz_m t_p - h_p\tau_p}, \tag{33}$$
$$\Lambda_{p,k} = -i\,e^{i\gamma_p}\,P_{\gamma_p}(w_k)\,e^{iw_k t_p + 2h_p\tau_p},$$

in which (28) reads

$$A_{q,p} = \sum_n \hat g_{q,n}V^{(1)}_{n,p}, \qquad B_{q,p} = \sum_n \hat g_{q,n}V^{(2)}_{n,p},$$
$$Y_p(Q) = \frac{1}{N_q}\sum_q A_{q,p}\,B_{q+Q,p},$$
$$X_Q(w_k) = \sum_p \varpi_p\,\Lambda_{p,k}\,Y_p(Q), \tag{34}$$
$$\chi_Q(w_k) = X_Q(w_k) + X_{-Q}(w_{k+N/2}),$$

since $-w_k = w_{k+N/2}$ by (2). The gauge cancels between the leaves and Λ. With $h_p = H_\lambda(\gamma_p)$ every term of the node sums is at most $\max_j|c_j G_q(z_j)|$, the entries of $V^{(1)}$ and $V^{(2)}$ are at most $\sum_j|c_j|$ and $\sum_m|d_m|$, and $|\Lambda_{p,k}| \le 1$ wherever $\mathrm{Im}(w_k e^{i\gamma_p}) \ge 2H_\lambda(\gamma_p)$; where this fails on part of an arc, a smaller h_p shares the remaining growth between the leaves and Λ. Since the leaves are bounded, the input $\hat g$ needs no weights $1/s_n$, unlike in the Cauchy route, and the modes with s_n below the target accuracy can be dropped (Appendix H). The matrices $V^{(1)}$, $V^{(2)}$ and Λ depend only on the geometry, the time rules and the partition. In the network of Fig. 6 the input train $\hat g$ is contracted with $V^{(1)}$ and $V^{(2)}$, the adder forms $Y_p(Q)$, a copy tensor multiplies by ϖ_p, and the Laplace train $\Lambda_{p,k}$ sends the time leg to the output points w_k. The result is the train $X_Q(w_k)$. The second term of (34) is the same train read at $-Q$ and $k + N/2$, that is, after the reflection $Q \to -Q$ of Appendix B and a NOT on the most significant bit of k, and the two trains are added as in Appendix K. The simplest partition takes $P_\gamma(w_k) = 1$ for the direction with the largest margin at w_k and 0 for the others.

III. TESTS

We test the network on the model (13) with $\lambda = 0.2$, $N = 512$ and $N_q = 1024$, at $\mu = 0.4$ and at $\mu = 0.25$. The second value lies below $\mu_{\mathrm{fwd}}(0.2) = 0.36$, so the copies $C_\lambda \pm w_k$ overlap C_λ and the Cauchy route of Appendix A fails, while $\lambda < \Lambda(\mu,\beta)$ still holds. The reference is the Lindhard function (F3) at all 512 points of C_μ and all transfer momenta. The tensors are built densely in double precision. Each direction has 8 Gauss-Legendre panels of 32 points at $\mu = 0.4$ and 16 panels at $\mu = 0.25$, with $d_a T_a = 30$, and the partition takes the direction with the largest margin. The gauge is $h_p = H_\lambda(\gamma_p)$ at $\mu = 0.4$; at $\mu = 0.25$, where $p_\mu < 2H_\lambda$, it is lowered to $h_p = 0.40$

TABLE I. The time network and the Cauchy route for the model (13) with $\lambda = 0.2$, $N = 512$ and $N_q = 1024$, compared with the Lindhard function at all points of C_μ and all transfer momenta. The error is $\max_{Q,k}|\chi - \chi_{\text{exact}}|/\max_{Q,k}|\chi_{\text{exact}}|$, and the largest entries of the trains are relative to $\max|\chi_{\text{exact}}|$. The bonds are the largest bonds of TT-SVDs of the dense trains at 10^{-10} with the criterion (I1); A, B and Y carry the time leg p (time network) or a Tucker leg (Cauchy route) at their first site. Real time alone uses $\gamma = 0$ on the upper half of C_μ and $\gamma = \pi$ on the lower half.

	time network		Cauchy route		real time alone
	$\mu = 0.4$	$\mu = 0.25$	$\mu = 0.4$	$\mu = 0.25$	$\mu = 0.4$
time points or Tucker ranks	1024	2048	$(31, 33, 44)$	$(67, 70, 103)$	512
smallest margin on C_μ	0.63	0.11			-0.52
error	4.6×10^{-14}	1.3×10^{-13}	1.7×10^{-7}	5.3×10^{-2}	0.11
largest entry of A, Y	4.6, 2.3	3.5, 1.7	47, 0.93		
rank of Λ over (p, k)	176	257			24
largest bond of A, B, Y	70, 68, 55	102, 100, 68	46, 47, 40		
largest bond of χ	40	62	38		

in real time and 3.15 in imaginary time. Table I collects the results. Its bonds are those of TT-SVDs of the dense tensors, while Table III gives those of the recompressed trains of the Cauchy route.

The time network gives the bubble to 4.6×10^{-14} of its maximum at $\mu = 0.4$ and 1.3×10^{-13} at $\mu = 0.25$, at every point of C_μ, the tips included; where $|\chi_Q|$ exceeds 10^{-3} of its maximum, the relative error is below 10^{-11}. The trains A, B, Y and X stay below five times the largest $|\chi_Q|$, and the entries of the leaves stay below 2 at $\mu = 0.4$ and below 44 at $\mu = 0.25$, while without the gauge the node sums reach 10^{39} and 10^{142}. The Cauchy route gives 1.7×10^{-7} at $\mu = 0.4$, the accuracy of its kernel compressed at 10^{-8}, and fails at $\mu = 0.25$. Real time alone, $\gamma = 0$ on the upper half and $\gamma = \pi$ on the lower half, is accurate on most of C_μ but errs by 0.11 of the maximum at the tips, where the integrand does not decay, and more time points do not reduce this error. The bound (26) quantifies why imaginary time is needed there. At $\mu = 0.4$ and $Q = \pi$, $c_2 = 3.75$, $L_\mu = 35.6$ and $R_\mu = 7.49$, so any formula entire in w errs by at least 0.088, that is 41 % of $\max_k|\chi_Q|$; at the other transfer momenta tested and at $\mu = 0.25$ the bound lies between 14 and 43 %.

The bonds of the result χ are those of the Cauchy route, 40 against 38 at $\mu = 0.4$, since the two routes represent the same tensor. The intermediate trains carry the time leg and are larger: the Laplace train has rank 176 over (p, k) at $\mu = 0.4$, against the 44 output modes of the Cauchy route, and the largest bonds of A and Y are 70 and 55 against 46 and 40. The core, however, is diagonal: the time leg sits at the first site of A, B and Y, where the Cauchy route carries its Tucker legs, and the adder step carries a leg of 1024 points instead of the leg (a, c) of B', of dimension $r_1 r_3 = 1364$. No HOSVD and no Cauchy sums over the nodes are needed. With the input on the window $s_n \geq 10^{-10}$, 131 modes, the error is 4.7×10^{-14}, and on $s_n \geq 10^{-6}$, 78 modes, it is 4.6×10^{-9} (1.5×10^{-8} at $\mu = 0.25$), so the input window follows the target accuracy. A smooth partition, with weights proportional to the square of the margin above $m_0 = 0.3$ or 0.5, gives the same accuracy with the longer windows $T_a \leq 57$ or 43 and a Laplace train of rank 205 or 187.

IV. TOWARD A GW SCHEME ON CONTOURS

The bubble is the first step of a GW scheme in which every frequency-dependent quantity, G, χ, the screened interaction W and the self-energy Σ, is stored on a contour, and every step is a tensor-network operation with a controlled error. This section gives the equations of such a loop, the conditions on its contours, the self-energy in the form of Sec. II C, and the order in which the pieces can be built and tested. The numbers come from the dense tests of Sec. IV E.

A. The loop

For electrons with two spin directions and an interaction v_Q, the polarization is $\Pi_Q = 2\chi_Q$, with χ_Q the bubble (14) of the current G, and the screened interaction and its screened part are

$$W_Q = \frac{v_Q}{1 - v_Q \Pi_Q}, \qquad \tilde{W}_Q = W_Q - v_Q = \frac{v_Q^2\, \Pi_Q}{1 - v_Q \Pi_Q}. \tag{35}$$

The self-energy is $\Sigma_q(z) = \Sigma^{\text{H}} + \Sigma^{\text{x}}_q + \Sigma^{\text{c}}_q(z)$. The Hartree term $\Sigma^{\text{H}} = v_0 \rho$ shifts the chemical potential, and the exchange and correlation terms are

$$\Sigma^{\text{x}}_q = -\frac{1}{N_q} \sum_Q v_Q\, \rho_{q-Q}, \qquad \Sigma^{\text{c}}_q(i\omega_n) = -\frac{1}{N_q \beta} \sum_{Q,m} G_{q-Q}(i\omega_n - i\nu_m)\, \tilde{W}_Q(i\nu_m), \tag{36}$$

where $\rho_q = \sum_j c_j G_q(z_j) = \int d\varepsilon\, f(\varepsilon) A_q(\varepsilon)$ is the occupation, a node sum with the weights (30) that needs no convergence factor, and $\rho = (2/N_q) \sum_q \rho_q$ is the density.

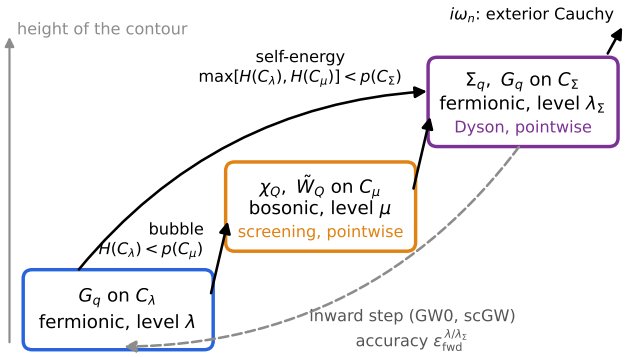


FIG. 7. The GW loop on contours. The boxes are placed at the heights of their contours, which rise along $G \to \chi \to \Sigma$. Each forward step, the bubble and the self-energy, needs the condition written next to it, while screening and Dyson's equation are pointwise. In G0W0 the chain ends on C_Σ, and the values at the Matsubara frequencies follow from the exterior Cauchy formula (A2). GW0 and scGW close the loop with the inward step (37) (dashed), whose accuracy is the forward accuracy raised to the level ratio.

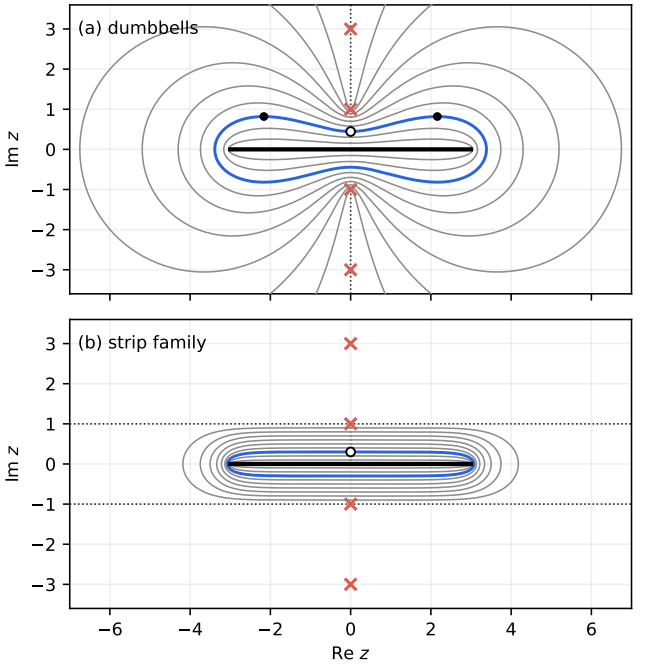


FIG. 8. The dumbbells (a) and the strip family (b) for $E = 3$ and $\beta = \pi$, at the levels $\lambda = 0.1, \ldots, 0.9$ (grey) with $\lambda = 0.3$ in blue, the band $[-3, 3]$ (black), the poles $\pm i$, $\pm 3i$ of f (crosses) and the level-1 curves (dotted): the imaginary axis beyond $\pm i$ for the dumbbells, the lines $\mathrm{Im}\, z = \pm 1$ for the strip family. The dots mark the highest points and the open circles the crossings of the imaginary axis. On the dumbbell at $\lambda = 0.3$ the top lies at 1.8 times the crossing; on the strip family the two coincide, $H = p = 0.30$.

Dyson's equation $G_q(z) = 1/[z - \varepsilon_q - \Sigma_q(z)]$, with ε_q measured from the chemical potential, closes the loop, and the chemical potential is fixed by ρ. In G0W0 the chain $G_0 \to \chi \to W \to \Sigma \to G$ runs once. In GW0 W is kept and Σ and G are iterated, and in self-consistent GW (scGW) the bubble is recomputed from the dressed G as well (Fig. 7).

B. Supports and contours

Each object is stored on a contour around its support. The support of G is a segment $[-E_G, E_G]$, the band at the start, which grows with the support of Σ. That of χ is $[-2E_G, 2E_G]$, and that of $\tilde{W}_Q$ is $[-\Omega_W, \Omega_W]$, which also covers the poles of W outside the particle-hole continuum, at the zeros of $1 - v_Q\Pi_Q$ on the real axis. On a finite lattice such a zero exists above every continuum. For an even band and $v_Q > 0$ its position ω satisfies $\omega^2 \le 4E_G^2 + 2v_Q c_2$, with c_2 of (25), and the coefficient of w^{-2} in $\tilde{W}_Q$, $2v_Q^2 c_2$, detects weight left outside the segment. The support of $\Sigma^{\rm c}$ is $[-E_G - \Omega_W, E_G + \Omega_W]$. In the iterated schemes G and Σ share one fermionic family on $[-E_G, E_G]$, with $E_G \ge E + \Omega_W$, so that Dyson's equation is pointwise, and χ and $\tilde{W}$ share one bosonic family, which encloses the pole of the Bose function at the origin and no other. On a finite lattice the pairs with $\varepsilon_{q+Q} = \varepsilon_q$ contribute $f'(\varepsilon_q)$ to $\Pi_Q(i\nu_0)$ and nothing at $\nu_m \neq 0$. The node sum $\chi_Q(0) = N_q^{-1} \sum_q \sum_j c_j G_q(z_j) G_{q+Q}(z_j)$ gives this Matsubara value, while a spectral representation of $\tilde{W}$ misses it.

Each forward step has one condition, the one of (18): the contours of its inputs lie below the point where the contour of its output crosses the imaginary axis. With $H(C)$ the height of a contour C and $p(C)$ the height of its crossing, the bubble needs $H(C_\lambda) < p(C_\mu)$, and the self-energy of Sec. IV C, computed on a contour C_Σ of the fermionic family at a level λ_Σ, needs $\max[H(C_\lambda), H(C_\mu)] < p(C_\Sigma)$. Screening, Dyson's equation, exchange and density are pointwise operations or node sums and need no condition. Since $H(C) \ge p(C)$, the levels rise along $G \to \chi \to \Sigma$. In G0W0 this costs nothing: Σ and G are known on C_Σ, and their values at the Matsubara frequencies, or anywhere outside C_Σ, follow from the exterior Cauchy formula (A2). In GW0 and scGW, Σ must be brought down from C_Σ to the contour C_λ of G. Within a family this is the level multiplier (H1), which multiplies the mode n by $e^{\alpha(\lambda_\Sigma - \lambda)n}$. If the forward steps give Σ on C_Σ to the accuracy $\varepsilon_{\rm fwd}$, the modes beyond $\ln(1/\varepsilon_{\rm fwd})/(\alpha\lambda_\Sigma)$ are below the forward error, and the step reaches

$$\varepsilon_{\rm in} \approx \varepsilon_{\rm fwd}^{\lambda/\lambda_\Sigma}. \tag{37}$$

The level ratio λ_Σ/λ is therefore the price of the loop.

On the dumbbells the top of a contour lies above its crossing, $H(C_\lambda) > p(C_\lambda)$, and the ratio H/p grows with E, so the chain needs a large level ratio. A second family has $H = p$ (Fig. 8). With $k = \tanh(\beta E/2)$, $k'^2 = 1 - k^2$,

and K and K' the complete elliptic integrals of parameters k^2 and k'^2, the map

$$z(\zeta) = \frac{2}{\beta}\,\mathrm{artanh}\left[k\,\mathrm{sn}\left(\frac{K}{2}(\zeta+1)\,\middle|\,k^2\right)\right] \tag{38}$$

sends the strip $\zeta = \phi + i\tau\lambda$ of Sec. I, with $\tau = 2K'/K$, onto the plane strip $|\mathrm{Im}\, z| < \pi/\beta$ cut along $[-E, E]$. The line $\lambda = 0$ goes to the segment, and the line $\lambda = 1$ to the lines $\mathrm{Im}\, z = \pm\pi/\beta$, which pass through the first poles $\pm i\pi/\beta$ of f; for a bosonic family the same map with $\beta/2$ gives lines through the first poles $\pm 2\pi i/\beta$ of the Bose function. The level curves of this map, the strip family, have their highest point on the imaginary axis, at

$$H_\lambda = p_\lambda = \frac{2}{\beta}\arctan\left[k\,\mathrm{sc}(K'\lambda\,|\,k'^2)\right], \tag{39}$$

with $\mathrm{sc} = \mathrm{sn}/\mathrm{cn}$, and their tips at $\pm(2/\beta)\,\mathrm{artanh}[k\,\mathrm{nd}(K'\lambda\,|\,k'^2)]$, with $\mathrm{nd} = 1/\mathrm{dn}$. The decay rate $\alpha = \pi\tau/4$ of their modes is smaller than that of the dumbbells, so at equal level and accuracy they need more nodes, one or two bits in quantics form. For $\beta E \gtrsim 15$, k is so close to 1 that the elliptic functions must be evaluated through the complementary nome.

C. The self-energy

The Bose function $n_B(s) = 1/(e^{\beta s} - 1)$ has the poles $i\nu_m$ with the residues $1/\beta$. Closing the bosonic sum in (36) with it on a contour C_μ of the bosonic family around the support of $\tilde{W}$, which encloses the pole at $s = 0$, and on the contour C_λ of G, both counterclockwise, gives

$$\begin{aligned}\Sigma^{\mathrm{c}}_q(w) = \frac{1}{N_q}\sum_Q \Big[&\frac{1}{2\pi i}\oint_{C_\mu} ds\, n_B(s)\,\tilde{W}_Q(s)\,G_{q-Q}(w-s)\\ &-\frac{\tilde{W}_Q(0)}{\beta}\,G_{q-Q}(w)\\ &+\frac{1}{2\pi i}\oint_{C_\lambda} dz\,[1-f(z)]\,G_{q-Q}(z)\,\tilde{W}_Q(w-z)\Big].\end{aligned} \tag{40}$$

The second term removes the residue of n_B at the origin, with the Matsubara value $\tilde{W}_Q(0)$. The third collects the residues at the cut of $G_{q-Q}(i\omega_n - s)$, where $n_B(i\omega_n - z) = -[1 - f(z)]$, and continues them to w. The formula holds when the shifted segments $w - [-\Omega_W, \Omega_W]$ lie outside C_λ and $w - [-E_G, E_G]$ outside C_μ, which on C_Σ is the condition $\max[H(C_\lambda), H(C_\mu)] < p(C_\Sigma)$. Shrinking both contours as in (7) gives, up to the static terms of a finite lattice, the spectral form

$$\begin{aligned}\Sigma^{\mathrm{c}}_q(w) = \frac{1}{N_q}\sum_Q \int d\varepsilon \int d\varepsilon'\, A_{q-Q}(\varepsilon)\,B_Q(\varepsilon')\\ \times\,\frac{1-f(\varepsilon)+n_B(\varepsilon')}{w-\varepsilon-\varepsilon'},\end{aligned} \tag{41}$$

with $B_Q(\varepsilon') = -\pi^{-1}\,\mathrm{Im}\,\tilde{W}_Q(\varepsilon' + i0^+)$.

The time representation (22) with $x = w - \varepsilon - \varepsilon'$ turns (41) into a product in time,

$$\begin{aligned}\Sigma^{\mathrm{c}}_q(w) = -\frac{i}{N_q}\sum_Q \int_0^{e^{i\gamma}\infty} dt\, e^{iwt}\\ \times\left[\bar{g}^{\mathrm{h}}_{q-Q}(t)\,\bar{u}_Q(t) + \bar{g}_{q-Q}(t)\,\bar{u}^{\mathrm{b}}_Q(t)\right],\end{aligned} \tag{42}$$

with $\bar{g}_q$ of (31) and the node sums

$$\begin{aligned}\bar{g}^{\mathrm{h}}_q(t) &= \sum_j (d_j - c_j)\,G_q(z_j)\,e^{-iz_j t},\\ \bar{u}_Q(t) &= \sum_k d^\mu_k\,\tilde{W}_Q(w_k)\,e^{-iw_k t},\\ \bar{u}^{\mathrm{b}}_Q(t) &= \sum_k d^\mu_k\,n_B(w_k)\,\tilde{W}_Q(w_k)\,e^{-iw_k t} - \frac{\tilde{W}_Q(0)}{\beta},\end{aligned} \tag{43}$$

where d^μ_k are the weights d of C_μ and $1 - f$ enters through $d_j - c_j$. The product is exact when $\mathrm{Im}(we^{i\gamma}) > (E_G + \Omega_W)|\sin\gamma|$, and by the argument of (24) it is stable when

$$\begin{aligned}\mathrm{Im}(we^{i\gamma}) &> H_\lambda(\gamma) + \Omega_W|\sin\gamma|,\\ \mathrm{Im}(we^{i\gamma}) &> H_\mu(\gamma) + E_G|\sin\gamma|,\end{aligned} \tag{44}$$

with $H_\lambda(\gamma)$ and $H_\mu(\gamma)$ the heights of C_λ and C_μ in the direction $ie^{-i\gamma}$. Both products are taken at $+t$, so the self-energy has no reflected term. The argument of (26) works here already at first order: $\Sigma^{\mathrm{c}}_q \simeq \Sigma_1/w$ at large w, with $\Sigma_1 = N_q^{-1}\sum_Q \int_0^\infty d\varepsilon'\, B_Q(\varepsilon')[1 + 2n_B(\varepsilon')] > 0$, so a formula entire in w errs on C_Σ by at least $2\pi\Sigma_1/L_\Sigma$, with L_Σ the length of C_Σ, and the self-energy needs the partition of (28) as well. Its network is that of Fig. 6 with two leaves per input, with the weights $d_j - c_j$ and d_j for G and d^μ_k and $d^\mu_k n_B(w_k)$ for $\tilde{W}$, a constant row $-\tilde{W}_Q(0)/\beta$ added to the second bosonic leaf, an adder that forms $(q - Q) + Q = q$, the two products summed on the time leg, and a Laplace train over eight directions. The diagonal directions serve the corners of flat contours, where the four of Sec. II C leave no margin.

D. Screening and Dyson's equation as trains

Screening and Dyson's equation are reciprocals of a train D over the points of a contour and the momenta, $D = 1 - v_Q\Pi_Q$ and $D = z - \varepsilon_q - \Sigma_q(z)$. The Newton iteration $x \leftarrow x(2 - Dx)$ squares the residual $1 - Dx$ at every step, so it converges from any start with $|1 - Dx_0| < 1$ at all points. The start $x_0 = D^*/\max|D|^2$ satisfies this, and the number of iterations grows like the logarithm of $\max|D|/\min|D|$. Cross interpolation of $1/D$ through the point evaluation of the train needs no iterations. The zeros of D lie on the real axis, at the poles of W and the quasiparticle poles of G, so $1/D$ is analytic in the same region as D, and its bonds stay close to those of D. The reciprocal fails when D has zeros near the contour, which happens for an attractive v_Q, where $1 - v_Q\Pi_Q(0)$ can

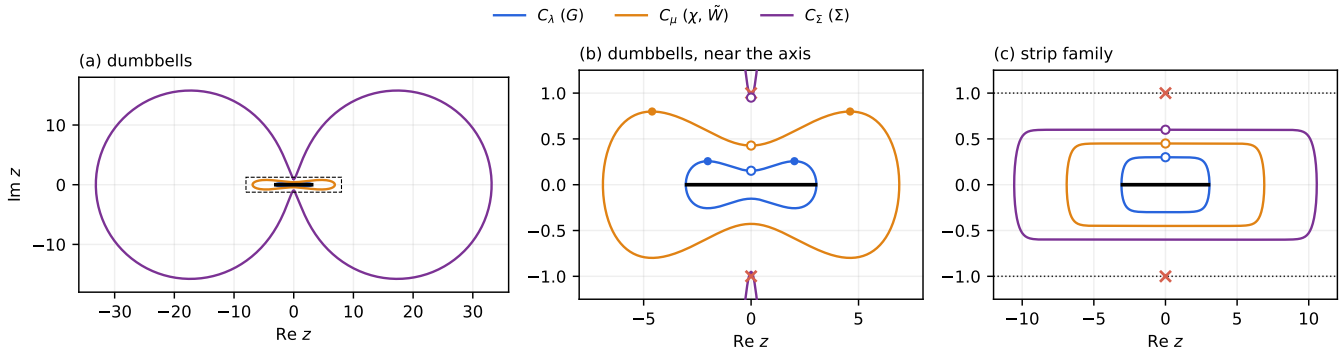


FIG. 9. The contours of the G0W0 test chains of Table II: C_λ of G (blue), C_μ of χ and $\tilde{W}$ (orange) and C_Σ of Σ (purple), with the band $[-3, 3]$ (black), the poles $\pm i$ of f (crosses), the crossings of the imaginary axis (open circles) and, where they lie in the frame, the highest points (dots). (a) Dumbbells at the levels 0.10, 0.14 and 0.80. C_Σ crosses the imaginary axis at 0.95 but reaches the height 15.8 and $|w| = 33$. (b) The dashed box of (a): the heights 0.26 and 0.80 of C_λ and C_μ lie below the crossings 0.43 and 0.95 of C_μ and C_Σ. (c) The strip family at the levels 0.30, 0.45 and 0.60, between the lines $\mathrm{Im}\, z = \pm 1$ (dotted), where every top is a crossing. Panels (b) and (c) are stretched vertically.

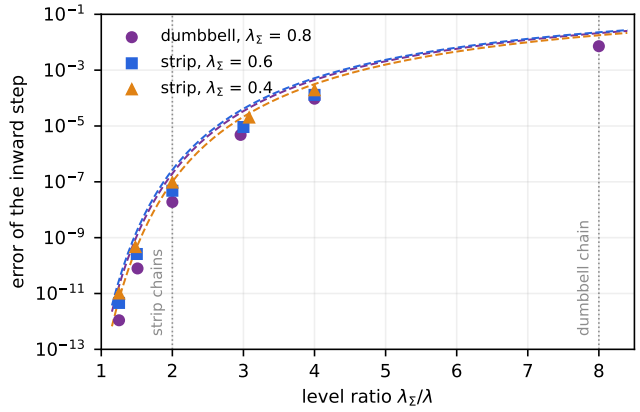


FIG. 10. The error of the inward step against the level ratio λ_Σ/λ, for Σ on the contours C_Σ of Table II (symbols), and the rule (37) with the forward errors of the three chains (dashed). The dumbbell chain needs the ratio 8, where the step gives 7×10^{-3}; the strip chains need 2, where it gives 5×10^{-8} and 9×10^{-8}.

vanish, or when W has poles outside the chosen segment; both show up in $\min|D|$ on the contour and in the sum rule of $\tilde{W}_Q$.

E. Tests and the order of the work

Dense tests on the chain (13) with $E = 3$, $\beta = \pi$, $N_q = 64$ and the interaction $v_Q = U + 2U_1 \cos Q$, $U = 2$, $U_1 = 0.5$, confirm the pieces. The zeros of $1 - v_Q \Pi_Q$ lie up to $\Omega_W = 6.22$, just above the continuum edge $2E$, and the families are built on $[-3, 3]$ for G_0, $[-6.72, 6.72]$ for χ and $\tilde{W}$, and $[-10.22, 10.22]$ for Σ. The closed form of G0W0 for this model, a sum over the poles of $\tilde{W}$ with the static terms, agrees with the direct Matsubara sum (36) to ten digits, and Table II compares the self-energy from (42) with it, for a chain of dumbbells and for two chains of the strip family, whose contours Fig. 9 shows. The self-energy is accurate to 10^{-13} in all three chains, its values at the Matsubara frequencies obtained from C_Σ by the exterior Cauchy formula agree with the closed form to 2×10^{-15}, and without the static term the error is 18 to 64 %. The bubble and $\tilde{W}$ on C_μ are accurate to 10^{-13}, and the density after Dyson's equation agrees with the Matsubara sum to 2×10^{-15}. The inward step follows (37) to within a factor of 2 above and 20 below (Fig. 10): from level 0.8 to 0.1 of the dumbbell chain it gives 7×10^{-3}, while from 0.6 to 0.3 of the first strip chain it gives 5×10^{-8}. The Newton reciprocals converge in 9 to 16 iterations, with $\max|D|/\min|D|$ from 3 to 5 for the screening and from 23 to 37 for Dyson's equation. On the first strip chain the widest bond at 10^{-10} is 62 for Π and for $\tilde{W}$, 63 for Σ and 74 for G.

The pieces can be built in the following order, each with its test.

1. The geometry for general E and β: the dumbbell and strip families, their heights, crossings and tips, and the choice of levels. Test: the node sum of $1/(z - a)$, the symmetries and the decay of the modes.

2. The self-energy network, dense and then as trains, reusing the bubble network: the two leaves per input, the convolution adder, the Laplace train over eight directions, the static term and the gauge. Test: the closed-form G0W0 to 10^{-13} and the Matsubara values by the exterior Cauchy formula. The time leg and the rank of the Laplace train set the cost; the compression of Appendix A 4 reduces them to the ranks of a Tucker core, at the accuracy of its truncation.

TABLE II. The G0W0 self-energy of the test model at all points of C_Σ, compared with its closed form. The chain of dumbbells uses the fermionic family $(3, \pi)$ for G_0, the bosonic family $(6.72, \pi/2)$ for χ and $\tilde{W}$, and the fermionic family $(10.22, \pi)$ for Σ; the strip chains use the strip family on the same segments with $\beta = \pi$ throughout. The margins are $p(C_\mu) - H(C_\lambda)$ and $p(C_\Sigma) - \max[H(C_\lambda), H(C_\mu)]$, and $\tilde{W}$ is the one computed from the bubble. The errors are the largest over all momenta and points relative to $\max|\Sigma^{\rm c}|$; the inward step brings Σ from C_Σ to the level of G within the family of Σ.

chain	levels of G, χ, Σ	margins	time points	error	without $-\tilde{W}_Q(0)/\beta$	inward step
dumbbells	0.10, 0.14, 0.80	0.17, 0.15	4352	6.3×10^{-14}	0.64	7.2×10^{-3}
strip	0.30, 0.45, 0.60	0.15, 0.15	9024	7.3×10^{-14}	0.20	4.8×10^{-8}
strip	0.20, 0.30, 0.40	0.10, 0.10	12096	1.3×10^{-14}	0.18	9.4×10^{-8}

3. Screening, Dyson's equation, exchange and chemical potential as trains, by Newton or cross interpolation. Test: the bonds of the reciprocals, their pointwise accuracy and the sum rule of $\tilde{W}$.

4. G0W0 at $N_q = 1024$ against a Matsubara GW code: $\Sigma(i\omega_n)$, the density, the energies, and the spectral function of Σ at the resolution set by $p(C_\Sigma)$.

5. GW0 with the inward step on the strip family at the level ratio 2: Σ brought to the level of G on the modes $n \le \ln(1/\varepsilon_{\rm fwd})/(\alpha\lambda_\Sigma)$, Dyson's equation there, and iteration. Test: a Matsubara GW0 code at the same β, and the convergence of ρ and of $\Sigma(i\omega_n)$. The accuracy of the inward step and its accumulation over the iterations are the main risk.

6. scGW: the bubble from the dressed G, the support E_G checked by the tail of the modes, and the poles of W tracked by the sum rule of $\tilde{W}$. Test: a Matsubara scGW code, the particle number and the f-sum rule.

The tests suggest the following choices.

1. Use the strip family for the iterated schemes, and the dumbbells for the bubble and for G0W0, where the forward steps suffice.

2. For the bosonic dumbbells use $\beta/2$, the doubled family of Sec. II B, whose ratio H/p on $[-6, 6]$ at level 0.2 is 1.7, against 3.2 at β; for the strip family use β.

3. Bring Σ inward, not G: Σ is smoother, and Dyson's equation at the lower level is pointwise.

4. Do not compress inside the loop. The inward step needs the forward accuracy of the diagonal core, which a Tucker core does not reach.

5. Keep the static terms of the finite lattice, at $Q = 0$ and from degenerate pairs; the node sums include them.

6. Heights scale like $1/\beta$ while the frequencies do not, so the time legs and the ranks grow like β. Test $\beta = 2\pi$ and 4π in step 4, before the loop.

7. Fix E_G at the start with a padded segment, and widen it only when the tail of the modes exceeds the tolerance.

8. Use margins of about 0.15, eight directions and a gauge that shares the growth between the leaves and the Laplace train.

Appendix A: The Cauchy route

In this appendix $G_{q\pm Q}$ at the shifted points $z \pm w$ is obtained from its values on C_λ with Cauchy's formula, instead of the time representation of Sec. II C. This route covers all of C_μ, its tips included, when the shifted points lie outside C_λ, that is $\lambda'_{\rm min} > \lambda$ in (4). Appendices B to E build its tensor network and test it. Appendix A 4 builds its kernel from the time route, which needs only $\lambda < \Lambda(\mu, \beta)$.

Formula (18) needs $G_{q\pm Q}$ at the shifted points $z \pm w$, off the contour, while the data are the values of G on the dumbbell C_λ, and the method evaluates G only there. Cauchy's formula, the first equality in (9), gives $G(x)$ at every x outside C_λ. For x *inside* C_λ the same integral gives 0 instead of $G(x)$: then $s \mapsto G(s)/(x-s)$ is analytic outside C_λ and $O(s^{-2})$, so its integral over C_λ equals its integral over C_R, which vanishes. With $x = z \pm w$ in (18), the bubble becomes a double integral over the dumbbell, in which G enters only on C_λ:

$$\chi_Q(w) = \frac{1}{N_q} \sum_q \frac{1}{(2\pi i)^2} \oint_{C_\lambda} dz \oint_{C_\lambda} ds\, f(z)\, G_q(z) \times \left[\frac{G_{q+Q}(s)}{z - s + w} + \frac{G_{q-Q}(s)}{z - s - w} \right]. \tag{A1}$$

1. The exterior-Cauchy form

The rule $\frac{1}{2\pi i} \oint_{C_\lambda} F\, dz \approx \sum_m d_m\, F(z_m)$ of Sec. II D turns the Cauchy formula in (9) into a sum over the nodes,

$$G(x) = \sum_{m=0}^{N-1} d_m \frac{G(z_m)}{x - z_m} \qquad (x \text{ outside } C_\lambda), \tag{A2}$$

up to the trapezoidal error. The two rules applied to (A1), c_j for the integral over z and d_m for the integral

over s, give the *exterior-Cauchy form* of the bubble,

$$\chi_Q(w_k) = \frac{1}{N_q}\sum_q \sum_{j,m} c_j\, G_q(z_j)\, d_m \times \left[\frac{G_{q+Q}(z_m)}{z_j - z_m + w_k} + \frac{G_{q-Q}(z_m)}{z_j - z_m - w_k}\right]. \tag{A3}$$

2. Fourier modes and the kernel

The output is expanded on C_μ in the same way as G in (32),

$$\chi_Q(w_k) = \sum_\ell \hat\chi_{Q,\ell}\, e^{i\pi\ell\phi_k/4}, \qquad \hat\chi_{Q,\ell} = \frac{1}{N}\sum_{k=0}^{N-1} \chi_Q(w_k)\, e^{-i\pi\ell\phi_k/4}. \tag{A4}$$

Insert (32) for $G_q(z_j)$ (index n') and for $G_{q\pm Q}(z_m)$ (index n) into (A3). The momentum average factorizes from the geometry:

$$\begin{aligned}\chi_Q(w_k) &= \sum_{n',n}\Big[P_Q^+(n',n)\, K_{n'n}(w_k) + P_Q^-(n',n)\, K_{n'n}(-w_k)\Big],\\ K_{n'n}(w) &= \sum_{j,m} c_j\, d_m\, \frac{e^{i\pi(n'\phi_j + n\phi_m)/4}}{z_j - z_m + w},\\ P_Q^\pm(n',n) &= \frac{1}{N_q}\sum_q \hat g_{q,n'}\hat g_{q\pm Q,n},\end{aligned} \tag{A5}$$

with the kernel $K_{n'n}$ and the momentum correlations $P_Q^\pm$. Since $\phi_{k+N/2} = \phi_k + 4$ and $\omega(\zeta+4) = -\omega(\zeta)$ (2), $-w_k = w_{k+N/2}$, so the second term is the first kernel read at $k + N/2$. Project on the output modes with (A4); for the second term substitute $k' = k + N/2$ and use $\phi_k = \phi_{k'} - 4$, which produces $e^{i\pi\ell} = (-1)^\ell$:

$$\begin{aligned}\hat\chi_{Q,\ell} &= \sum_{n',n} \widehat K_{n'n\ell}\big[P_Q^+(n',n) + (-1)^\ell P_Q^-(n',n)\big],\\ P_Q^\pm(n',n) &= \frac{1}{N_q}\sum_q \hat g_{q,n'}\,\hat g_{q\pm Q,n},\end{aligned} \tag{A6}$$

$$\widehat K_{n'n\ell} = \frac{1}{N}\sum_k e^{-i\pi\ell\phi_k/4} \times \sum_{j,m} c_j\, d_m\, \frac{e^{i\pi(n'\phi_j + n\phi_m)/4}}{z_j - z_m + w_k}. \tag{A7}$$

This is the exterior-Cauchy form (A3), where the kernel depends only on the two dumbbells and on β (through c_j); all of G, model and momentum, is in $P_Q^\pm$. For each k the inner double sum is a two-dimensional FFT over (j,m) of the array $c_j d_m/(z_j - z_m + w_k)$, and the outer sum is an FFT over k. In exact arithmetic the sums in (A6) run over all N values of n', n and ℓ. In practice only a window of each index matters, because the coefficients of G and of χ decay exponentially (Appendix H).

3. Compressing the kernel

The kernel (A7) is compressed into Tucker form by a truncated higher-order SVD (HOSVD). For a tensor $T_{n'n\ell}$ let $T_{(i)}$ be its mode-i unfolding (the matrix with the i-th index as rows and the other two, combined, as columns). The HOSVD takes for $U^{(i)}$ the leading left singular vectors of $T_{(i)}$ and forms the core $C = T \times_1 U^{(1)\dagger} \times_2 U^{(2)\dagger} \times_3 U^{(3)\dagger}$, i.e.

$$\begin{aligned}C_{abc} &= \sum_{n'n\ell} T_{n'n\ell}\, \overline{U^{(1)}_{n'a}}\,\overline{U^{(2)}_{nb}}\,\overline{U^{(3)}_{\ell c}},\\ T_{n'n\ell} &\approx \sum_{a=1}^{r_1}\sum_{b=1}^{r_2}\sum_{c=1}^{r_3} C_{abc}\, U^{(1)}_{n'a}\, U^{(2)}_{nb}\, U^{(3)}_{\ell c}.\end{aligned} \tag{A8}$$

Each rank r_i is the smallest with $\sum_{s>r_i}(\sigma_s^{(i)})^2 \le \frac{\varepsilon^2}{3}\|T\|_F^2$, $\sigma_s^{(i)}$ the singular values of $T_{(i)}$. Since the three truncations are orthogonal projections, their errors add at most in quadrature, $\|T - \hat T\|_F^2 \le \sum_i \sum_{s>r_i}(\sigma_s^{(i)})^2 \le \varepsilon^2\|T\|_F^2$, so ε bounds the relative error of the compressed tensor. The result is quasi-optimal: its error is at most $\sqrt{3}$ times that of the best Tucker approximation with the same ranks. Applied to the weighted kernel $\widetilde K_{n'n\ell} = s_{n'} s_n \widehat K_{n'n\ell}$ of Appendix H, whose weights s_n make a relative truncation error of $\widetilde K$ an error of the same size in $\hat\chi$,

$$s_{n'}\, s_n\, \widehat K_{n'n\ell} \approx \sum_{a,b,c} C_{abc}\, U^{(1)}_{n'a}\, U^{(2)}_{nb}\, U^{(3)}_{\ell c}, \tag{A9}$$

at $\varepsilon = 10^{-8}$ the ranks are $(r_1, r_2, r_3) = (31, 33, 44)$ for $(\lambda,\mu) = (0.2, 0.4)$ and $(56, 60, 87)$ for $(0.1, 0.2)$. The core has $r_1 r_2 r_3 = 45\,012$ entries in the example.

Inserting (A9) into (A6) and absorbing the weights into the leaves, $\tilde U^{(i)}_{na} = U^{(i)}_{na}/s_n$, gives the bubble in factorized form, the formula the network computes:

$$\begin{aligned}A_{q,a} &= \sum_n \hat g_{q,n}\, \tilde U^{(1)}_{na}, \qquad B_{q,b} = \sum_n \hat g_{q,n}\, \tilde U^{(2)}_{nb},\\ Y_c^\pm(Q) &= \frac{1}{N_q}\sum_q \sum_{a,b} A_{q,a}\, C_{abc}\, B_{q\pm Q,b},\\ \hat\chi_{Q,\ell} &= \sum_c U^{(3)}_{\ell c}\Big[Y_c^+(Q) + (-1)^\ell\, Y_c^-(Q)\Big],\\ \chi_Q(w_k) &= \sum_\ell \hat\chi_{Q,\ell}\, e^{i\pi\ell\phi_k/4}.\end{aligned} \tag{A10}$$

Per momentum the input is reduced to $r_1 + r_2$ numbers; $Y^\pm$ are $r_1 r_2$ cyclic correlations in q contracted with the

core; the output needs r_3 modes. Nothing in (A10) depends on N except the final synthesis. For any G, the definition of $Y^{\pm}$ gives the exact identity

$$Y_c^-(Q) = Y_c^+(-Q \bmod N_q), \tag{A11}$$

since $B_{q-Q,b} = B_{q+(-Q),b}$. The branch Y^- therefore follows from Y^+ by the reflection $Q \to -Q$, without a second contraction (Appendix D), and since the reflection changes a bond by at most a factor 2, the bonds of the two branches agree within that factor. For the even band $\varepsilon_q = 3\cos q$ the two branches are equal: $A_{-q,a} = A_{q,a}$ and $B_{-q,b} = B_{q,b}$, and the substitution $q \to -q$, which maps the momentum grid onto itself, gives

$$\frac{1}{N_q}\sum_q A_{q,a}B_{q-Q,b} = \frac{1}{N_q}\sum_q A_{q,a}B_{q+Q,b}, \tag{A12}$$

that is, $Y^-(Q) = Y^+(Q)$ exactly on the grid. Then $\hat{\chi}_{Q,\ell} = [1+(-1)^\ell]\sum_c U^{(3)}_{\ell c} Y_c^+(Q)$: the odd modes vanish, which is the evenness $\chi_Q(-w) = \chi_Q(w)$ of the bubble of an even band. Computed by two separate zip-ups, each recompressed at 10^{-9}, the two trains differ by $\|Y^- - Y^+\|/\|Y^+\| = 1.6\times10^{-11}$. The network does not use this symmetry: it computes both branches, as it must for a band that is not even.

The ranks are small because G and χ enter only through Cauchy transforms. Equation (19) writes the bubble as the average of $[f(\varepsilon) - f(\varepsilon')]/(w+\varepsilon-\varepsilon')$ with the weights $A_q(\varepsilon)A_{q+Q}(\varepsilon')$: a trilinear form in fA_q, A_{q+Q} and the output kernel $1/(w+\varepsilon-\varepsilon')$, and (A3) is its discretization. The first G enters only through $\sum_j c_j G_q(z_j)/(z_j - x) = \int f(\varepsilon)A_q(\varepsilon)\,d\varepsilon/(\varepsilon - x)$, the Cauchy transform of fA_q at the points $x = z_m - w_k$; the second through the Cauchy transform of $A_{q\pm Q}$ at $z_j \pm w_k$; and the output is the Cauchy transform of a two-particle spectral function on $[-2E, 2E]$, evaluated on C_μ. So each leg of $\widetilde{K}$ is a Cauchy kernel $1/(x-\varepsilon)$ between a segment and a set at a positive distance from it, with $\Lambda(\mu,\beta)$ the lowest level of the cloud $[-E,E] \pm C_\mu$ (Sec. II B):

leg	source segment	target set	lowest level of the target
1 (n')	$[-E,E]$, weight fA	$[-E,E] - C_\mu$ (the "cloud")	$\Lambda(\mu,\beta)$
2 (n)	$[-E,E]$, weight A	$[-E,E] + C_\mu$ (the same set)	$\Lambda(\mu,\beta)$
3 (ℓ)	$[-2E,2E]$ (doubled family, level 0)	C_μ	μ

In dumbbell coordinates a Cauchy kernel from the band to a contour at level Λ is the diagonal multiplier $e^{-\alpha\Lambda n}$ of (H1), so its singular values decay exponentially at the rate $\alpha\Lambda$ and its ε-rank is $\log(1/\varepsilon)/(\alpha\Lambda)$. The discrete Lehmann representation works the same way, with the dumbbell level in the role of $\log\Lambda_{\mathrm{DLR}}$. Measured for μ from 0.2 to 0.8, the ranks follow, within 3 %,

$$r_3 \approx 0.90\,\frac{\log(1/\varepsilon)}{\alpha\mu}, \qquad r_{1,2} \approx 0.65\,\frac{\log(1/\varepsilon)}{\alpha\Lambda(\mu,\beta)}; \tag{A13}$$

for $\varepsilon = 10^{-8}$ this gives 42 and 31 at $\mu = 0.4$ (measured 44 and 31), and 84 and 53 at $\mu = 0.2$ (measured 87 and 56). The prefactors are below 1 because the HOSVD measures the error in the Frobenius norm over the whole cloud, most of which lies far above level Λ; the sup-norm version (the number of poles needed to represent a Green function on C_Λ) is $\log(1/\varepsilon)/(\alpha\Lambda)$ within 1 to 3 %. The ranks therefore grow like $\log(1/\varepsilon)$ (about +9 per two decades for $(0.2, 0.4)$ and +16 for $(0.1, 0.2)$), like $1/\mu$, and roughly linearly with β at fixed level, because at fixed level the whole dumbbell family moves toward the real axis as $T \to 0$. They do not depend on N or on λ.

4. The kernel from the time route

The tensors (33) give the kernel of this route in canonical form, with the time leg as its rank,

$$K^{\mathrm{t}}_{n'nk} = \sum_p \varpi_p\, V^{(1)}_{n',p}\, V^{(2)}_{n,p}\, \Lambda_{p,k}. \tag{A14}$$

By (34), $X_Q(w_k) = \sum_{n',n} P^+_Q(n',n)\, K^{\mathrm{t}}_{n'nk}$, which is the first term of the bubble in Appendix A 2 with $K^{\mathrm{t}}_{n'nk}$ in place of $K_{n'n}(w_k)$, and the second term is $X_{-Q}(w_{k+N/2})$, read at $k + N/2$ as there. When the copies $C_\lambda \pm w_k$ lie outside C_λ, the two kernels agree up to the quadrature errors. When they overlap C_λ, $K_{n'n}(w_k)$ no longer gives the bubble, while K^{t} still does for every G analytic off the band, because it comes from (21) applied to products of exponentials. Projected on the output modes with (A4) and weighted by $s_{n'}s_n$, K^{t} takes the place of $\widehat{K}$ in (A9), and the network of Appendices B to E runs unchanged on all of C_μ whenever $\lambda < \Lambda(\mu,\beta)$.

For $N_q = 64$ and $(\lambda,\mu) = (0.2, 0.4)$, K^{t} agrees with $K_{n'n}(w_k)$ on the window of modes to 2.2×10^{-11} of its largest entry, its HOSVD at 10^{-8} has the ranks $(31, 33, 44)$ of Appendix A 3, and the network with this tree gives the bubble to 1.7×10^{-7}. For $(0.2, 0.25)$, below $\mu_{\mathrm{fwd}}(0.2)$, the two kernels differ by 36 % of the largest entry and the tree of the Cauchy kernel gives the bubble only to 5.3×10^{-2}, while the tree of K^{t} has the ranks $(46, 49, 70)$ and gives 5.0×10^{-7}.

The canonical form also compresses the time leg of Sec. II D. Let $U^{(3)}$ hold the leading left singular vectors of the unfolding of the weighted K^{t} over its output index k. Replacing Λ by $\Lambda U^{(3)}U^{(3)\dagger}$ reduces the output leg to r_3 at the accuracy of the truncation: at $(0.2, 0.4)$, $r_3 = 44$ and the error 1.6×10^{-7} at 10^{-8}, and $r_3 = 55$ and 2.1×10^{-9} at 10^{-10}; at $(0.2, 0.25)$, 70 and 4.1×10^{-7}, and 88 and 7.1×10^{-9}. The time network can therefore trade its accuracy for the ranks of the Cauchy route.

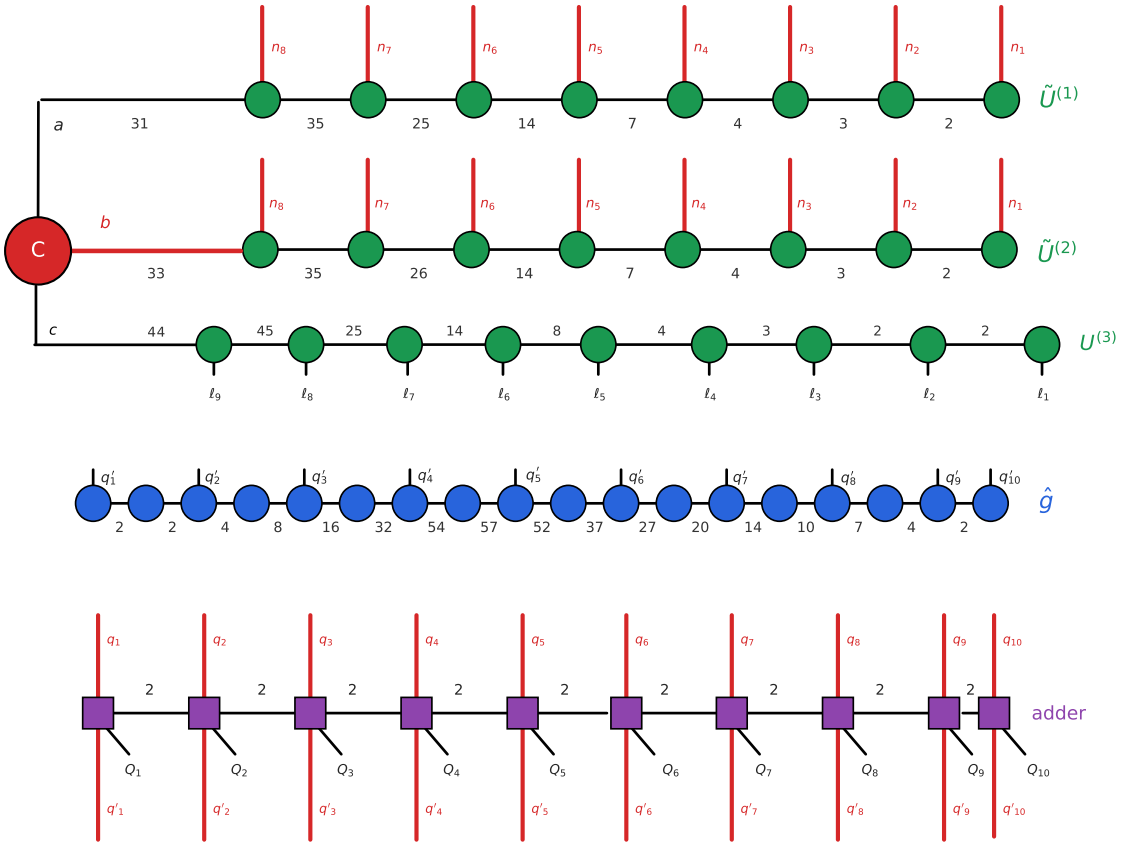


FIG. 11. The ingredients, with the bond dimensions of the run (tolerances in Appendix C). Top: the Tucker tree of the weighted kernel, the core C with the legs a, b, c and the leaves $\tilde{U}^{(1)}$, $\tilde{U}^{(2)}$ over the bits $n_8 \dots n_1$ and $U^{(3)}$ over $\ell_9 \dots \ell_1$. Middle: the input $\hat{g}$, whose sites alternate between momentum bits and bits of n. Bottom: the adder, with the input legs q_i, q'_i and the output legs Q_i; its bond is the carry.

Appendix B: Quantics tensor trains

An index $q \in \{0, \dots, 2^R - 1\}$ is written in binary,

$$q = \sum_{i=1}^{R} q_i \, 2^{R-i}, \qquad q_i \in \{0, 1\}, \tag{B1}$$

with q_1 the most significant bit and q_R the least significant one. A vector $v(q)$ of length 2^R is thereby a tensor $v(q_1, \dots, q_R)$ with R indices of dimension $d = 2$, its *quantics* form; d is the *physical dimension* of every site. A signed index such as a Fourier mode $n \in [-2^{R-1}, 2^{R-1})$ is stored in *two's complement*, as the bits of n mod 2^R:

$$n = -n_1 \, 2^{R-1} + \sum_{i=2}^{R} n_i \, 2^{R-i}, \tag{B2}$$

so n_1 is the sign bit, and the nonnegative modes come first, followed by the negative ones, each $n < 0$ at position $2^R + n$. This is the order in which an FFT returns its modes. "Most significant first" means the chain carries $q_1, q_2, \dots, q_R$ in this order; "least significant first" the reverse.

A tensor train (TT, or matrix-product state, MPS) writes a tensor with L indices as a product of matrices, one per index (*site*),

$$\begin{aligned} v(\sigma_1, \dots, \sigma_L) &= M^{[1]}_{\sigma_1} M^{[2]}_{\sigma_2} \cdots M^{[L]}_{\sigma_L}, \\ M^{[i]}_{\sigma_i} &\in \mathbb{C}^{\chi_{i-1} \times \chi_i}, \qquad \chi_0 = \chi_L = 1. \end{aligned} \tag{B3}$$

The χ_i are the *bond dimensions* ("bonds"); the train stores $\sum_i d\, \chi_{i-1} \chi_i$ numbers instead of d^L. A tensor with an additional dense index (the Tucker indices a, b, c below) is a train whose first or last core carries that index, so that χ_0 or χ_L equals its dimension. The smallest bond χ_i that represents the tensor exactly depends only on the tensor and the site order: it is the rank of the *unfolding* matrix whose rows are indexed by $(\sigma_1, \dots, \sigma_i)$ and whose columns by $(\sigma_{i+1}, \dots, \sigma_L)$. A matrix-product operator (MPO) is the same object for a linear map, $W(\sigma'_1 \dots \sigma'_L; \sigma_1 \dots \sigma_L) = W^{[1]}_{\sigma'_1 \sigma_1} \cdots W^{[L]}_{\sigma'_L \sigma_L}$, with cores carrying an input and an output bit. Applying an MPO of bond D_W to a train of bond χ gives, before any compression, a train of bond $D_W \chi$.

For a function of two variables the bit strings are concatenated, and the order matters because the bonds are ranks of unfoldings. *Sequential* order puts all bits of one

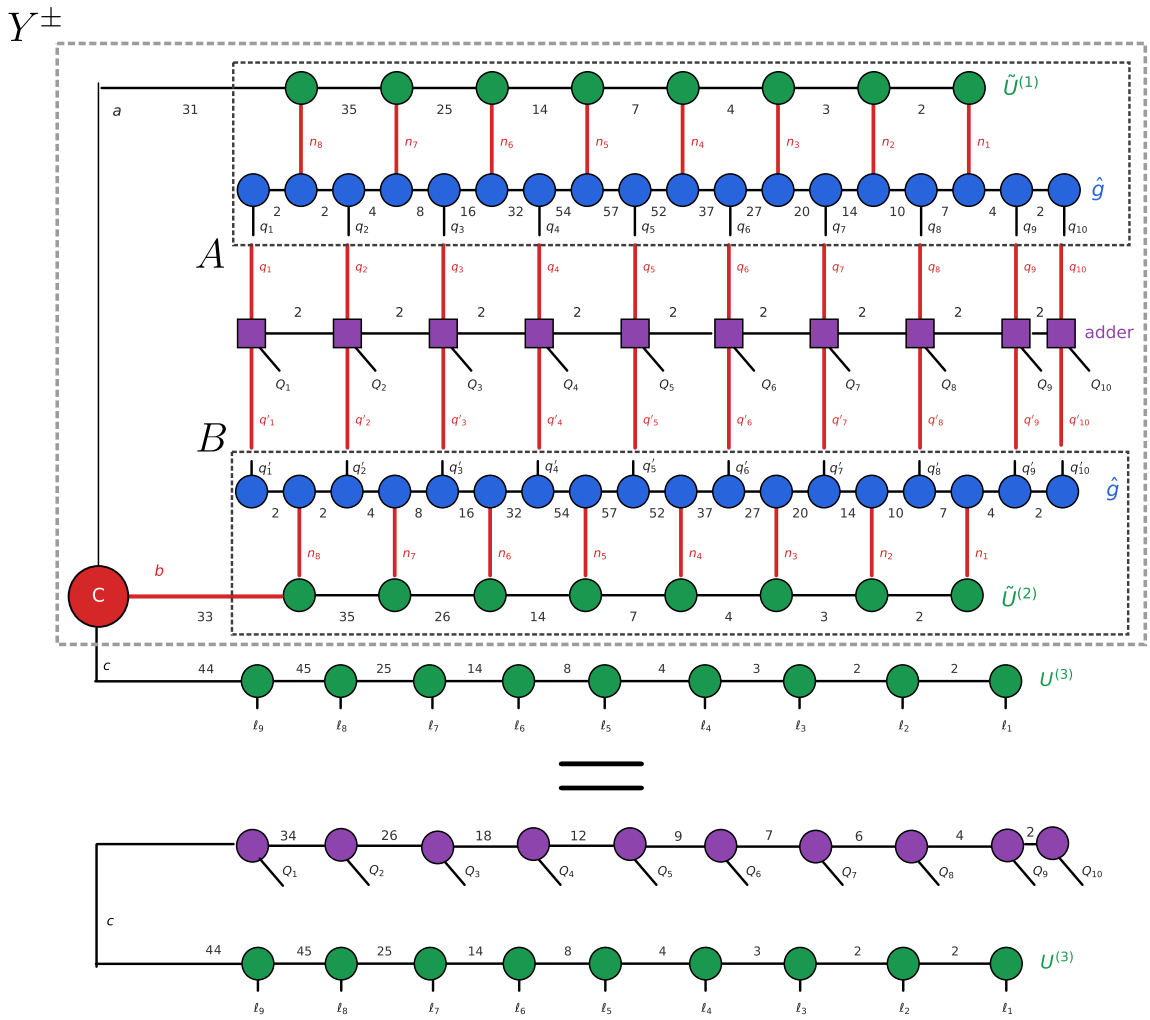


FIG. 12. The network, with the bond dimensions of the run; red legs are contracted. The dashed box A holds $\hat{g}$ and the leaf $\tilde{U}^{(1)}$, contracted over the legs n_i in step ①, and the box B holds $\hat{g}$ and $\tilde{U}^{(2)}$, contracted in step ②. The grey dashed box is the network of step ④, whose result is the train $Y^{\pm}(Q;c)$: A and B joined by the adder over the legs q_i, q'_i, with the core C on their legs a and b. The output leaf $U^{(3)}$ is attached through c. Below the equals sign is the same network after step ④. The drawing shows Y^{+}. For Y^{-} the adder is replaced by the subtractor ($q' = q - Q$, bond: the borrow) and $U^{(3)}$ carries the sign $(-1)^{\ell} = (-1)^{\ell_9}$; the bonds are the same. By (A11), the Y^{-} network equals the Y^{+} network read at $-Q$.

variable before those of the other ($n_1 \dots n_R\, q_1 \dots q_{R_q}$); *interleaved* order alternates them ($q_1 n_1 q_2 n_2 \dots$), possibly with one variable reversed ($q_1 n_R q_2 n_{R-1} \dots$). Bits that interact should sit next to each other. A smooth function of two variables is best sequential: the single matrix coupling them has low rank, and interleaving spreads it over many bonds. A Fourier-type coupling $e^{2\pi i\, nq/2^R}$ is best interleaved with one variable reversed, because the bit q_i couples only to the bits n_j with $i + j > R$ (Appendix J). For this reason the input $\hat{g}_{q,n}$ of the method is interleaved, with n least significant first, and the output $\hat{\chi}_{Q,\ell}$ is sequential (Appendix M).

The cyclic correlation $\frac{1}{N_q}\sum_q A_q B_{q+Q}$ over the momentum grid, $N_q = 2^{R_q}$, is the contraction of $A(q)$ and $B(q')$ with the tensor $\delta_{q',\,(q+Q) \bmod N_q}$, the *adder*, which is an MPO of bond dimension 2 over the bits of q, q' and Q. Binary addition propagates a carry $c \in \{0,1\}$ from the least significant bit upward; the local tensor at bit i is

$$\big[W^{[i]}\big]_{c_{\rm in}c_{\rm out}}(q_i, Q_i, q'_i) = \delta\big(q'_i,\ q_i \oplus Q_i \oplus c_{\rm in}\big) \times \delta\big(c_{\rm out},\ \lfloor (q_i + Q_i + c_{\rm in})/2 \rfloor\big), \tag{B4}$$

with $\oplus$ the sum modulo 2, $c_{\rm in} = 0$ at the least significant bit and the carry out of the most significant bit discarded (addition modulo N_q: momenta are periodic). The bond index is the carry, so the bond is exactly $D_W = 2$ at every site, for any R_q and with no truncation. The *subtractor* $\delta_{q',\,(q-Q) \bmod N_q}$ has the same form with a borrow, $q'_i = q_i \oplus Q_i \oplus b_{\rm in}$ and $b_{\rm out} = 1$ iff $q_i - Q_i - b_{\rm in} < 0$; equivalently, it is the adder with the roles of q and q' exchanged ($q = q' + Q$). In the network (Appendix D) the legs q_i and q'_i are contracted with the two input branches and the legs

Q_i remain open, so one contraction gives the correlation for all N_q transfer momenta at once.

The reflection $Q \to -Q \bmod N_q$ (two's-complement negation: complement every bit, then add 1) is also an MPO of bond 2: a bitwise NOT followed by the "+1" tensor $P[c_{\text{out}}, \text{out}, \text{in}, c_{\text{in}}] = 1$ iff $\text{out} = \text{in} \oplus c_{\text{in}}$ and $c_{\text{out}} = \text{in} \wedge c_{\text{in}}$, with $c_{\text{in}} = 1$ at the least significant bit and the final carry discarded.

A train produced by an operation (an MPO applied to a train, a sum as in Appendix K, a contraction) in general has bonds larger than needed. A QR sweep followed by a sweep of truncated SVDs, the *recompression*, brings them down to what the tolerance requires, with the error bound of TT-SVD (Appendix L). Every intermediate of the contraction is recompressed this way at relative tolerance 10^{-9}.

To apply an MPO of bond D_W to a train of bond χ, one can form the product train (bond $D_W\chi$) and recompress it, at a cost $O(Ld\,(D_W\chi)^3)$ that is prohibitive when $D_W\chi$ reaches 10^4 (it would be $\approx 13\,800$ in step ④ below). The *zip-up* contracts and truncates site by site in a single sweep. At site i the carried matrix, whose rows are the truncated bond χ'_{i-1}, is contracted with the i-th MPO and train cores, which produces a matrix with $d\chi'_{i-1}$ rows (the truncated bond and the output leg of site i) and $D_W\chi$ columns (the right bonds of the MPO and of the train); a truncated SVD factors it, U becomes the finished core i, and $SV^\dagger$ is carried to site $i+1$. The product of bond $D_W\chi$ is never formed: the largest matrix has one side of size $d\chi'$ and the other $D_W\chi$ (the largest SVD of the run is 550×5832). Since the truncation at site i does not yet know the sites to its right, it is slightly suboptimal; a final recompression sweep removes the excess (the two rows of bonds in Table III). The same scheme contracts two trains over shared physical legs: in steps ① and ② the leaf train is zipped along $\hat{g}$ over the legs n_i, with an SVD after every q site, and in step ④ the adder sits between A and B', the legs q_i, q'_i are summed and Q_i stays open.

Appendix C: The tensors

The numbers in this and the following sections are for $N_q = 1024$ momenta and $N = 512$ contour points, that is $R_q = 10$ and $R = 9$ bits, with $\lambda = 0.2$ and $\mu = 0.4$. In Fig. 11 and below, all bits are written with the most significant bit first unless stated otherwise; n and ℓ are in two's complement, i.e. FFT order (B2), and bond dimensions are listed from left to right. The input $\hat{g}_{q,n}$ is the raw array (32) as a train; Appendix M explains why the raw array is truncated and the weight $1/s_n$ is put into the leaves. The leaves are the Tucker factors of (A9) written as trains over the bits of n or ℓ, with the Tucker index attached at the least significant bit, where the leaf bonds stay below the Tucker rank. Every tensor is computed to a relative tolerance ε, in the sense of Appendix A 3 for the HOSVD and of Appendix I for the trains:

tensor	computed by	ε
Tucker tree: core C, factors $U^{(i)}$	HOSVD of the weighted kernel	10^{-8}
input $\hat{g}_{q,n}$	TT-SVD of the raw array	10^{-6}
leaves $\tilde{U}^{(1)}$, $\tilde{U}^{(2)}$, $U^{(3)}$	TT-SVD	10^{-10}
inverse DFT F	TT-SVD of the MPO	10^{-12}
adder, subtractor	exact, bond 2	none
intermediates of the contraction	zip-up and recompression	10^{-9}

Appendix D: The contraction

Step ① contracts $\hat{g}$ with the leaf $\tilde{U}^{(1)}$ over the legs $n_1 \dots n_8$, and step ② contracts it with $\tilde{U}^{(2)}$; these are the dashed boxes A and B of Fig. 12, with the contracted legs in red. The core C is then contracted with the first site of B over b (step ③), which gives B'. The grey dashed box is the network of step ④: A and B' joined by the adder (for Y^+) or the subtractor (for Y^-) over the legs q_i and q'_i, with only c and $Q_1 \dots Q_{10}$ left open. Its result is the train $Y^\pm(Q; c)$, drawn below the equals sign, still attached to the output leaf $U^{(3)}$ through c. Step ⑤ does this last contraction for both signs, with the sign $(-1)^\ell$ on ℓ_9 for Y^-, and adds the two branches. Step ⑥, the inverse DFT $\ell \to k$ (Appendix J), is not drawn. Figure 13 shows the recompressed trains A, B and $Y^\pm$, and Table III gives the bonds and the cost of every step.

With both zip-ups of step ④, 29 s each, the contraction takes about 60 s. The largest bond after recompression is 84, in B; from the core on, every bond is at most 34, the bond of the result itself. Three objects are never formed: the weighted input $\hat{g}/s$, which is incompressible; the correlation $M(Q; a, b)$, whose leg has dimension 1023; and the uncompressed products inside the zip-ups, whose bond would reach $\approx 13\,800$ in step ④.

The branch Y^- does not need its own zip-up. By (A11), $Y^-(Q) = Y^+(-Q \bmod N_q)$: the subtractor version of Fig. 12 is the adder version read at the reflected momentum. The reflection $Q \to -Q$ is the bitwise NOT on every Q bit followed by the exact "+1" MPO of bond 2 (Appendix B), applied to the train $Y^+(Q; c)$. The reflection takes 0.01 s, and the reflected Y^+ agrees with Y^- from the subtractor zip-up to 1.7×10^{-10}. We therefore recommend one zip-up for Y^+ (29 s) followed by the reflection; the whole contraction then takes about 31 s instead of 60 s.

Appendix E: Accuracy

Compared with the Lindhard bubble (F3) at all 1024 transfer momenta and all 512 points of C_μ, the network

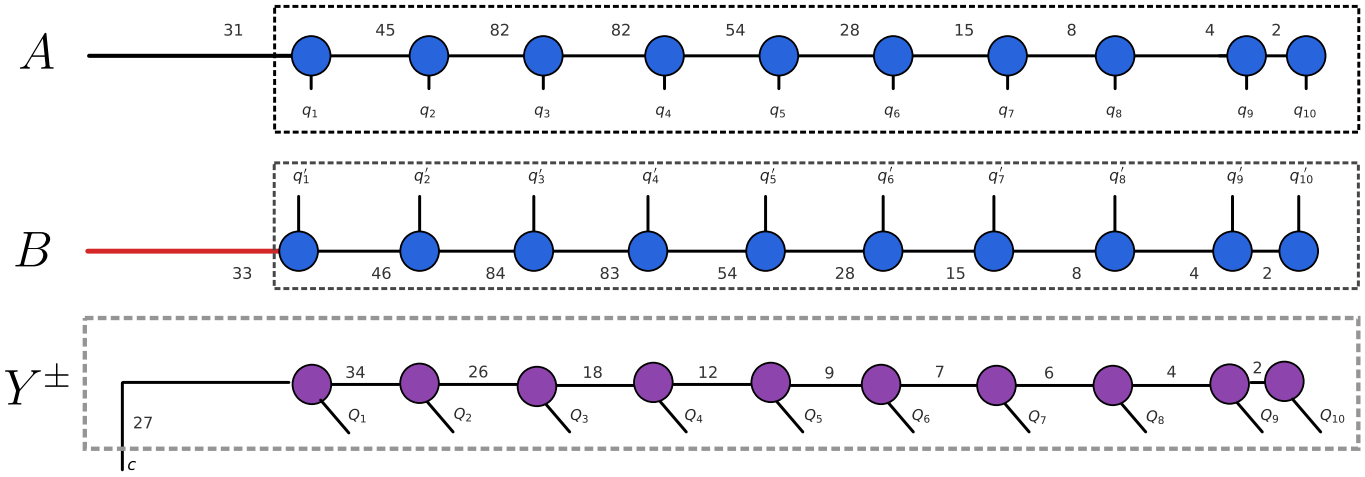


FIG. 13. The intermediate trains after recompression, with the bond dimensions of the run: $A(q;a)$ from step ①, $B(q';b)$ from step ②, and $Y^{\pm}(Q;c)$ from step ④, the same for both signs. Their bonds are the second rows of Table III.

TABLE III. The contraction, step by step. For every step the bonds of the result are given before recompression (first row) and after it (second row). The first, second and fourth steps are zip-ups (Appendix B), so their bonds before recompression are those left by the zip-up. In the cost column χ is the bond dimension of every train and the dimension of the Tucker legs a, b, c; $d = 2$ is the physical dimension of the legs q_i, q'_i, Q_i, n_i, ℓ_i, k_i; $D_W = 2$ is the bond of the adder; $D = 11$ is the bond dimension of the DFT MPO; L is the number of sites.

step	contract	over	result	bonds before recompression → after recompression	cost
①	$\hat{g}$ with $\tilde{U}^{(1)}$	$n_1 \dots n_8$ (zip-up, SVD after every q site)	A: a, $q_1 \dots q_{10}$	31, 50, 100, 193, 293, 139, 47, 10, 4, 2 → 31, 45, 82, 82, 54, 28, 15, 8, 4, 2	$O(Ld^2\chi^4)$
②	$\hat{g}$ with $\tilde{U}^{(2)}$	$n_1 \dots n_8$ (zip-up)	B: b, $q_1 \dots q_{10}$	33, 52, 104, 196, 293, 139, 47, 10, 4, 2 → 33, 46, 84, 83, 54, 28, 15, 8, 4, 2	$O(Ld^2\chi^4)$
③	C with the first site of B	b	B': (a, c), $q_1 \dots q_{10}$	unchanged, no recompression	$O(\chi^4)$
④	A, adder (or subtractor), B'	a, q_i, q'_i (zip-up, SVD after every site)	$Y^{\pm}$: c, $Q_1 \dots Q_{10}$	44, 88, 173, 275, 211, 101, 46, 24, 13, 7 → 27, 34, 26, 18, 12, 9, 7, 6, 4, 2	$O(LD_W d^2\chi^4)$ for each sign
⑤	$U^{(3)}$ with Y^+ and with Y^-, then add	c; sign $(-1)^{\ell}$ on ℓ_9	$\hat{\chi}$: $\ell_1 \dots \ell_9$, $Q_1 \dots Q_{10}$	4, 4, 6, 8, 16, 28, 50, 90, 54, 68, 52, 36, 24, 18, 14, 12, 8, 4 → 2, 2, 2, 3, 5, 9, 16, 24, 24, 34, 26, 18, 12, 9, 7, 6, 4, 2	$O(Ld\chi^3)$
⑥	F with $\hat{\chi}$	$\ell_1 \dots \ell_9$	χ: $k_9 \dots k_1$, $Q_1 \dots Q_{10}$	4, 8, 16, 33, 55, 72, 64, 48, 24, 34, 26, 18, 12, 9, 7, 6, 4, 2 → 2, 4, 7, 8, 11, 16, 20, 24, 24, 34, 26, 18, 12, 9, 7, 6, 4, 2	$O(LdD^3\chi^3)$

TABLE IV. Error per transfer momentum $Q = 2\pi m/1024$.

m	error	$\max_k\lvert\chi_{\text{exact}}\rvert$
1	2×10^{-4}	3×10^{-5}
16	8×10^{-7}	6×10^{-3}
64	5×10^{-8}	0.05
160	9×10^{-9}	0.10
256	5×10^{-9}	0.13
512	1.7×10^{-7}	0.22

gives

$$\max_{Q,k}|\chi - \chi_{\text{exact}}| \Big/ \max_{Q,k}|\chi_{\text{exact}}| = 1.7 \times 10^{-7}. \qquad \text{(E1)}$$

Table IV gives the error per transfer momentum $Q = 2\pi m/1024$, $\max_k|\chi - \chi_{\text{exact}}|/\max_k|\chi_{\text{exact}}|$. The absolute error is about 10^{-8} at every Q; the relative error grows at small Q only because χ_Q itself vanishes there. The contraction adds no error above 10^{-9}: the MPS result

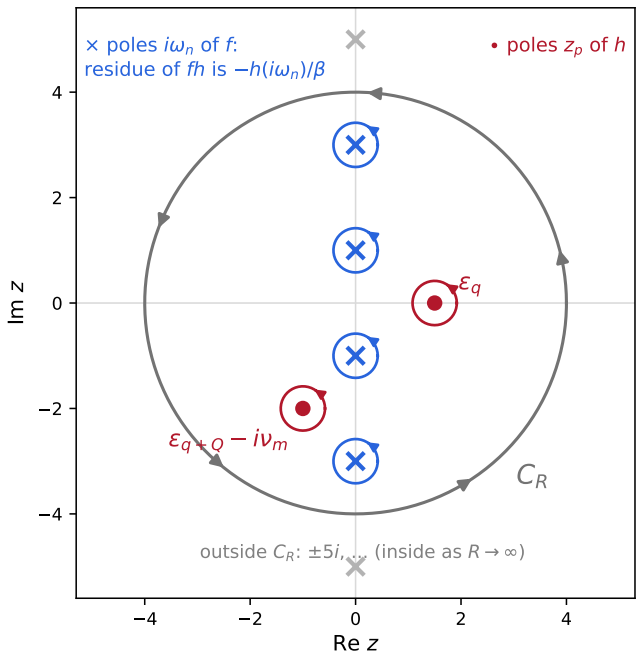


FIG. 14. The residue argument behind (F1), for $\beta = \pi$ and the h of the model with $m = 1$. Crosses: the poles $i\omega_n = \pm i, \pm 3i, \ldots$ of f, where fh has the residues $-h(i\omega_n)/\beta$. Dots: the poles of h, ε_q (here 1.5) and $\varepsilon_{q+Q} - i\nu_m$ (here $-1 - 2i$). The integral over the circle C_R equals the sum of the integrals over the small circles, i.e. $2\pi i$ times all residues inside C_R. It tends to zero as $R \to \infty$, so the residues at the crosses and at the dots add up to zero.

and the same factors contracted densely agree to that level. The error comes from the compressions, mainly that of the kernel at 10^{-8}; by Appendix H, a relative truncation of the weighted kernel is directly an error of $\hat{\chi}$.

Appendix F: The Lindhard function

For the model (13), $h(z) = G_q(z)\, G_{q+Q}(z + i\nu_m) = 1/[(z - \varepsilon_q)(z + i\nu_m - \varepsilon_{q+Q})]$ has only two poles, and the residues at the Matsubara frequencies in (16) can be traded for the residues at these two poles (Fig. 14).

Let h be meromorphic and $O(z^{-2})$ at infinity, with poles z_p none of which is a Matsubara frequency, and let C_R be the circle $|z| = R$, counterclockwise, with R an even multiple of π/β, halfway between two poles of f. By the residue theorem, $\frac{1}{2\pi i}\oint_{C_R} fh\, dz$ is the sum of the residues of fh inside C_R: those at the poles $i\omega_n$ of f and those at the poles z_p of h. On C_R, $|h| \le c/R^2$ and $|f|$ stays bounded ($f \to 1$ for $\mathrm{Re}\, z \to -\infty$, $f \to 0$ for $\mathrm{Re}\, z \to +\infty$, and C_R passes between the poles), so the integral is at most of order $2\pi R \cdot c/R^2 \to 0$. As $R \to \infty$ every pole ends up inside, so the two groups of residues add up to zero, and with (16)

$$\frac{1}{\beta}\sum_n h(i\omega_n) = \sum_{\text{poles } z_p \text{ of } h} \operatorname*{Res}_{z=z_p} \big[f(z)\, h(z)\big]. \tag{F1}$$

For example, $h(z) = 1/[(z - a)(z - b)]$ has the residues $f(a)/(a - b)$ at a and $f(b)/(b - a)$ at b, so

$$\frac{1}{\beta}\sum_n \frac{1}{(i\omega_n - a)(i\omega_n - b)} = \frac{f(a) - f(b)}{a - b}\,: \tag{F2}$$

an infinite sum, whose terms fall like $1/n^2$, replaced by two terms. With $a = 1.5$, $b = -1 - 2i$ and $\beta = \pi$ the right-hand side is $-0.23163 + 0.18530\, i$, and the sum over $|n| < 10^6$ reaches it to 2×10^{-7}. The bubble of the model is this example with $a = \varepsilon_q$ and $b = \varepsilon_{q+Q} - i\nu_m$. With $f(\varepsilon_{q+Q} - i\nu_m) = f(\varepsilon_{q+Q})$ it is the Lindhard function

$$\chi_Q(i\nu_m) = \frac{1}{N_q}\sum_q \frac{f(\varepsilon_q) - f(\varepsilon_{q+Q})}{i\nu_m + \varepsilon_q - \varepsilon_{q+Q}}. \tag{F3}$$

The right-hand side is a rational function of $i\nu_m$; replacing $i\nu_m$ by any $w \notin \mathbb{R}$ gives the analytic continuation of χ_Q off the real axis, as (18) does for a general G (Sec. II B).

Appendix G: Convergence of the dumbbell quadrature

Let Φ be 8-periodic and analytic in the strip $|\mathrm{Im}\, \phi| < a$, and bounded there by M. Shifting the line of integration in the Fourier coefficients $\hat{\Phi}_n = \frac{1}{8}\int_{-1}^{7} \Phi\, e^{-i\pi n\phi/4} d\phi$ to $\mathrm{Im}\, \phi = \mp a$ (the vertical sides cancel by periodicity) gives $|\hat{\Phi}_n| \le M e^{-\pi|n|a/4}$. The N-point trapezoidal sum of $\Phi = \sum_n \hat{\Phi}_n e^{i\pi n\phi/4}$ is exact for $|n| < N$ and aliases the modes $n = pN$ onto the mean, so

$$\begin{gathered}\Delta\phi \sum_j \Phi(\phi_j) - \int_{-1}^{7} \Phi\, d\phi = 8 \sum_{p \neq 0} \hat{\Phi}_{pN}, \\ \Big|\ \cdot\ \Big| \le \frac{16 M\, e^{-\pi N a/4}}{1 - e^{-\pi N a/4}}.\end{gathered} \tag{G1}$$

For the integrands of (30), $\Phi(\phi) = J(\zeta)F(\omega(\zeta))f(\omega(\zeta))$ with $\zeta = \phi + i\tau\lambda$. A complex shift $\phi \to \phi + i\tau\delta$ moves the level from λ to $\lambda + \delta$, so Φ is analytic in the strip $|\mathrm{Im}\, \phi| < a$ with $a = \tau d$, where d is the distance in level from λ to the nearest singularity: the band (level 0, $d = \lambda$), the poles of f (level 1, $d = 1 - \lambda$), or a singularity of F. With $\alpha = \pi\tau/4$, the error is

$$\varepsilon_{\text{trap}} \approx e^{-\alpha N d}. \tag{G2}$$

For the integral over z in (18), $F = G_q(z)[G_{q+Q}(z + w) + G_{q-Q}(z - w)]$, and the singularities of F outside the band are the shifted cuts, at level distance $\approx \lambda'_{\min}$ (near the band the level is locally proportional to the height, so the

level of $z_j \pm w_k$ above the band equals the level distance of z_j from the shifted cut), hence $d = \min(\lambda, \lambda'_{\min})$. The estimate $e^{-\alpha N \min(\lambda,\lambda'_{\min})}$ matches the measured errors: for $\mu = 0.2$, $\lambda = 0.12$ and $N = 2^8$ it gives $e^{-28.6} = 4 \times 10^{-13}$, and $\lambda = 0.2$, $N = 2^8$ gives $e^{-7.8} = 4 \times 10^{-4}$. The two distances are balanced at $\lambda \approx \mu/2$, so the levels are chosen in that ratio.

Appendix H: Decay of the coefficients and the weighted kernel

The function $g(\zeta) = G_q(\omega(\zeta))$ is 8-periodic in $\operatorname{Re}\zeta$ and analytic in the open strip $0 < \operatorname{Im}\zeta < 2\tau$, i.e. $0 < \lambda < 2$: ω maps the strip into $\mathbb{C} \setminus [-E, E]$, where G_q is analytic, and at the two poles of ω (at $\zeta = 1 + i\tau$ and $5 + i\tau$) $G_q(\omega) \to 0$ is still analytic. The lines $\lambda = 0$ and $\lambda = 2$ are the boundary, where ω is on the band. The coefficients at level λ are $\hat{g}_n(\lambda) = \frac{1}{8}\int_{-1}^{7} g(\phi + i\tau\lambda) e^{-i\pi n\phi/4} d\phi$ (the continuum version of (32)). Shifting the line of integration from height $\tau\lambda_0$ to $\tau\lambda$ inside the strip is allowed by Cauchy's theorem (the vertical sides cancel by periodicity) and gives

$$\begin{aligned} &\hat{g}_n(\lambda) = e^{-\alpha(\lambda-\lambda_0)n}\, \hat{g}_n(\lambda_0), \\ &0 < \lambda, \lambda_0 < 2, \qquad \alpha = \frac{\pi\tau}{4}. \end{aligned} \tag{H1}$$

Changing the level is a diagonal operation on the Fourier modes. Because $\hat{g}_n(\lambda_0)$ stays bounded as $\lambda_0 \to 0^+$ and as $\lambda_0 \to 2^-$ (the boundary values of G on the band are bounded or integrable),

$$\begin{aligned} &|\hat{g}_n(\lambda)| \lesssim e^{-\alpha\lambda n} \qquad (n > 0), \\ &|\hat{g}_n(\lambda)| \lesssim e^{-\alpha(2-\lambda)|n|} \quad (n < 0)\,: \end{aligned} \tag{H2}$$

positive modes "see" the band at level 0, negative modes its image at level 2. At $\lambda = 0.2$ the rates are 0.196 and 1.77 per mode, so the coefficients reach 10^{-15} of their maximum at $n \approx 176$ and $n \approx -20$; smaller coefficients are rounding noise of the double-precision data. The same holds for χ on C_μ with μ in place of λ, which sets the ℓ window.

The worst case among all admissible G is a pole on the band, whose coefficients saturate (H2). Define the *weights*

$$s_n = \begin{cases} e^{-\alpha\lambda n}, & n \geq 0, \\ e^{-\alpha(2-\lambda)|n|}, & n < 0, \end{cases} \tag{H3}$$

and write the bubble as $\hat{\chi}_{Q,\ell} = \sum_{n'n} \widetilde{K}_{n'n\ell} \widetilde{P}^{\pm}_{Q}(n', n)$ with the *weighted kernel* $\widetilde{K}_{n'n\ell} = s_{n'} s_n \widehat{K}_{n'n\ell}$ and $\widetilde{P} = P/(s_{n'} s_n)$. For every G whose coefficients are bounded by the worst case, $|\hat{g}_{q,n}| \leq c\, s_n$, the entries of $\widetilde{P}$ are bounded by c^2, so an error $\delta\widetilde{K}$ of the weighted kernel gives $|\delta\hat{\chi}_{Q,\ell}| \leq c^2 \sum_{n'n} |\delta\widetilde{K}_{n'n\ell}|$. A relative truncation error ε of $\widetilde{K}$ is therefore directly an error $\approx \varepsilon$ of $\hat{\chi}$, uniformly over the admissible G. Compressing the unweighted $\widehat{K}$ would spend rank on the high modes, which never carry signal, and would give no error control. The input window of Appendix A 2 ends where s_n falls below 10^{-8}. The modes left out cannot change $\hat{\chi}$ at the tolerance of the kernel: their rows in $\widetilde{K}$ are at most 3.3×10^{-12} of $\|\widetilde{K}\|_F$. Keeping them would only add noise. In double precision the HOSVD returns their rows of $U^{(i)}$ as rounding noise of size about $2 \times 10^{-16} \|\widetilde{K}\|_F / \sigma^{(i)}_{r_i} \approx 3 \times 10^{-8}$, and the factor $1/s_n$ in the leaves turns it into entries of order 10^7 on the window $[-32, 159]$, against at most 1.2×10^3 on $[-10, 93]$. On the wider window the error is the same, 1.7×10^{-7}, but the largest bonds of A and B grow from 82 and 84 to 103 and 102, and step (4) takes 45 s instead of 29 s. On the window no floor is needed in (H3).

The weighted kernel does not depend on λ. By (H1), the ratios $a_{q,n} \equiv \hat{g}_{q,n}(\lambda)/s_n(\lambda)$ are the coefficients at level 0 for $n \geq 0$ and at level 2 for $n < 0$: level-free Laurent coefficients of G_q, independent of λ. The bubble is $\hat{\chi}_{Q,\ell} = \sum_{n'n} \widetilde{K}_{n'n\ell} \frac{1}{N_q} \sum_q [a_{q,n'} a_{q+Q,n} + (-1)^\ell a_{q,n'} a_{q-Q,n}]$, and neither $\hat{\chi}$ (a contour integral, independent of the contour) nor the a's depend on λ. Hence $\widetilde{K}$ is the same trilinear form for every forward-safe λ, up to the quadrature error $e^{-\alpha N(\lambda'_{\min}-\lambda)}$. Measured at $\mu = 0.4$, the weighted kernels built at $\lambda = 0.2$ and $\lambda = 0.1$ (both $N = 512$) differ by 1.7×10^{-11} of their maximum. Consequently the ranks of the compressed kernel depend on μ, β and ε only, not on λ or N, and the kernel can be built at the cheapest (λ, N) and applied to data at another λ. A small λ makes the margin $\lambda'_{\min} - \lambda$ large, so N is set by the window alone, with the window cut at the target ε. For $\mu = 0.4$ this means $\lambda = 0.1$, $N = 256$ and 2 s instead of 10 s; for $\mu = 0.2$, $\lambda = 0.05$, $N = 512$ and 29 s plus 19 s of HOSVD instead of 108 s plus 120 s, with identical ranks and errors.

Appendix I: TT-SVD and the truncation criterion

Any tensor can be brought into the form (B3) by successive SVDs: reshape v into the $d \times d^{L-1}$ matrix $v_{(1)}$, compute $v_{(1)} = USV^\dagger$, take $M^{[1]} = U$ (its columns are the first bond), reshape $SV^\dagger$ into a $(d\chi_1) \times d^{L-2}$ matrix, and continue. At each bond the singular values are truncated by the *relative Frobenius criterion*: keep the smallest χ_i with

$$\sum_{s > \chi_i} \sigma_s^2 \leq \frac{\varepsilon^2}{L-1} \|v\|^2, \tag{I1}$$

i.e. each bond discards at most the fraction $\varepsilon^2/(L-1)$ of the squared norm. The truncations at different bonds are orthogonal projections, so the discarded weights add and the relative error in Frobenius norm is at most ε; the result is within a factor $\sqrt{L-1}$ of the best train with the same bonds. All tolerances quoted for trains are relative

tolerances ε in this sense. The "bond" or "rank" of an object in the tables is the bond dimension it needs at the stated ε.

Appendix J: The discrete Fourier transform as an MPO

For $k = \sum_i k_i 2^{R-i}$ and $\ell = \sum_j \ell_j 2^{R-j}$,

$$\begin{aligned} e^{2\pi i k\ell/2^R} &= \prod_{i,j} \exp\left(2\pi i\, k_i \ell_j\, 2^{R-i-j}\right) \\ &= \prod_{i+j>R} \exp\Big(\frac{2\pi i\, k_i \ell_j}{2^{\,i+j-R}}\Big), \end{aligned} \tag{J1}$$

because the factors with $i + j \le R$ equal 1. The most significant bit k_1 therefore couples only to the least significant bit ℓ_R, k_2 to ℓ_{R-1} and ℓ_R, and so on. If site i of the MPO carries the pair (k_i, ℓ_{R+1-i}), i.e. one index in reversed bit order, then k_i couples to the pairs at sites $i' \le i$ with the phase $2\pi/2^{\,i-i'+1}$, which decreases geometrically with the distance along the chain and is negligible beyond a distance of about $\log_2(1/\varepsilon)$. The interaction is quasi-local and the MPO has bond dimension $O(\log(1/\varepsilon))$. This is the quantum-Fourier-transform circuit, and the reason the QFT returns the modes in bit-reversed order. With both indices in the same order the strongest coupling, $k_1\ell_R$, joins the two ends of the chain and the bond is maximal. The inverse transform of the method, $F(k,\ell) = e^{i\pi\ell\phi_k/4} = e^{-i\pi\ell/4} e^{2\pi i\ell k/N}$, is a phase times (J1); with ℓ most significant first and k least significant first its MPO has bonds $2, 4, 8, 11, 11, 8, 4, 2$ at tolerance 10^{-12}, so its bond dimension is $D = 11$.

Appendix K: Sum of two trains

The sum $v + v'$ of two trains is the train with cores

$$\begin{aligned} M_\sigma^{[1]} \oplus M_\sigma'^{[1]} &= \begin{pmatrix} M_\sigma^{[1]} & M_\sigma'^{[1]} \end{pmatrix}, \\ M_\sigma^{[L]} \oplus M_\sigma'^{[L]} &= \begin{pmatrix} M_\sigma^{[L]} \\ M_\sigma'^{[L]} \end{pmatrix}, \\ M_\sigma^{[i]} \oplus M_\sigma'^{[i]} &= \begin{pmatrix} M_\sigma^{[i]} & 0 \\ 0 & M_\sigma'^{[i]} \end{pmatrix} \quad (1 < i < L), \end{aligned} \tag{K1}$$

a direct sum: the bonds add, $\chi_i + \chi_i'$, and are brought back down by recompression. A factor that acts on one site, such as the sign $(-1)^\ell = (-1)^{\ell_R}$ (a diagonal tensor on the least significant bit of ℓ), is multiplied into the corresponding core before the sum.

Appendix L: Recompression

A train is recompressed in two sweeps. (i) A QR sweep from left to right: reshape the first core into a $(d\chi_0) \times \chi_1$ matrix, factor it as $M^{[1]} = QR$ with Q having orthonormal columns, keep Q as the new core and multiply R into the core to its right; repeat up to the last site. Now every core but the last is left-orthogonal, and the norm of the train sits in the last core. (ii) A sweep from right to left with truncated SVDs: factor the current core (reshaped to $\chi_{i-1} \times (d\chi_i)$) as $USV^\dagger$, truncate by (I1), keep $V^\dagger$ as the new core and multiply US into the core to its left. Because the rest of the train is orthogonal, the singular values of the local matrix are those of the full unfolding at that bond, so each truncation is optimal for its bond, and with (I1) the total relative error is at most ε, as for TT-SVD. The cost is $O(Ld\chi^3)$ for a train of bond χ.

Appendix M: Design choices

1. Truncate the raw $\hat{g}$ and put the weight $1/s_n$ into the leaves. The raw coefficients decay as in (H2), and truncating them at 10^{-6} still gives a bubble accurate to 10^{-9}: the leaves divided by s_n lie in the range of the unweighted kernel, which is bounded, so errors in the high modes are not amplified. The weighted array $\hat{g}/s$ compresses poorly: at 10^{-6} its bond is 99, close to the 104 modes of the window, against 57 for the raw $\hat{g}$. In it every mode of the window has the same size, and for each momentum $\hat{g}_n/s_n$ is close to a plane wave $e^{-i\pi n\phi(\varepsilon_q)/4}$ whose frequency moves nonlinearly with q.

2. Interleave the bits of the input and keep the output sequential (Appendix B, bit orders). For $\hat{g}$, interleaving q and n with n least significant first gives bond 57 at 10^{-6}; all n bits first gives 75, and all q bits first 145. For the output, frequency bits first and then Q gives 25 to 32, while interleaving gives 36 to 63 for $\hat{\chi}$ and 48 to 83 for χ (at 10^{-6} to 10^{-8}). The contraction delivers the output in the best order by construction.

3. Contract the core into B before the adder. Correlating A and B first creates $M(Q; a, b)$, with a front leg of 1023; its zip-up swells to bonds of 600 to 2600, and capping them costs 3 to 13 % accuracy.

4. Put ℓ on the same bits as k. The inverse DFT is then the square, QFT-type transform (J1), with bond at most 11. Keeping only the 128 window modes on 7 bits gives a rectangular map with bond 32.

5. Compute Y^- by reflecting Y^+ (Appendix D): one zip-up instead of two, 31 s instead of 60 s.

6. In the Cauchy route, choose $\lambda \lesssim \mu/2$, so that the copies $C_\lambda \pm w_k$ stay outside C_λ ($\lambda'_{\min} > \lambda$, Sec. I A), keep the input modes with $s_n \ge 10^{-8}$ (Appendix H), and set the tolerances to 10^{-8} for the kernel, 10^{-6} for the input, 10^{-10} for the leaves, 10^{-12} for the DFT and 10^{-9} for the recompression. The kernel ranks grow like $\log(1/\varepsilon)$ and like $1/\mu$

(and with β), as in (A13). They do not grow with N or λ, and the kernel may be built at a smaller N than the data's (Appendix H).

7. In the time network, put the four directions on one time leg. The second term of (34) needs Y at the negated times, which in real time is the complex conjugate of Y and in imaginary time a separate grid, so filling part of C_μ with the symmetries $\chi_Q(\bar{w}) = \chi_Q(w)^*$ and $\chi_Q(-w) = \chi_{-Q}(w)$ saves at most one of the four grids. The symmetry maps of the output, $k \to N/2-k$ with complex conjugation and $k \to k+N/2$ with $Q \to -Q$, are exact MPOs of bond 2 and serve as checks.

8. Keep the direction fixed on each arc of C_μ. The direction with the largest margin at each point raises the smallest margin from 0.11 to 0.22 at $(\lambda, \mu) = (0.2, 0.25)$ and leaves it at 0.63 at $(0.2, 0.4)$, but the leaves then depend on the output point: on a quarter of C_μ the leaf $V_{n,p,k}$ has rank 35 over k at 10^{-10}, and the time leg of that quarter grows from 512 to 8960.